Part A - Project Summary

Call EUI-IA Call 4

A.1 Project Identification

Acronym Twin4forestLife

Title Twin4forestLife: A municipal Digital Twin for safe, resilient and climate-adapted urban-forest territories

Project Number EUI04-1006

(Main) Urban Authority Munipality of Castrovillari

ERDF rate 80%

Topic Security, safety and preparedness

Project Duration

Initiation Phase

Start Date 01/12/2026

End Date 28/02/2027

Implementation Phase

Start Date 01/03/2027

End Date 28/02/2029

Duration in months 24

Closure Phase

Start Date 01/03/2029

End Date 31/05/2029

A.2 Project Summary

Description

Twin4forestLife addresses the increasing wildfire and climate-risk exposure of small and medium-sized urban authorities at the interface between urban areas, forests, agricultural land, river corridors and tourism-sensitive landscapes. In Castrovillari and Cassano all'Ionio, existing wildfire-risk maps, civil-protection plans, regional AIB data, emergency areas, access routes and communication procedures remain fragmented and not yet usable as real-time municipal decision-support tools. The project will test a municipal Digital Twin connecting regional wildfire intelligence with local data on vegetation, climate conditions, evacuation routes, firefighting water logistics, vulnerable assets and risk communication needs. The solution will be piloted through two living labs: Castrovillari will focus on the urban-forest interface and prevention-oriented vegetation management; Cassano all'Ionio on wildfire-interface preparedness, emergency access and smart communication. The innovation lies in making the regional wildfire-monitoring and Digital Twin ecosystem actionable at municipal level. The project will not duplicate regional AIB, Control Room or firefighting systems; it will provide the missing local operational layer for prevention, preparedness and community engagement. Twin4forestLife will strengthen local innovation capacity by transforming fragmented data into actionable municipal decisions for safer, more resilient and climate-adapted territories.

Partner	Country	ERDF co-financing (maximum 80%)		Contribution (minimum 20%)			Total 100% TEC (total eligible costs)	
		EUR	ERDF rate	Public	Private	Total	Budget	% of project budget
PP 1 - Munipality of Castrovillari	IT	420,000.00	80.00%	105,000.00	0.00	105,000.00	525,000.00	21.00%
PP 2 - Municipality of Cassano allo Ionio	IT	320,000.00	80.00%	80,000.00	0.00	80,000.00	400,000.00	16.00%

Partner	Country	ERDF co-financing (maximum 80%)		Contribution (minimum 20%)			Total 100% TEC (total eligible costs)	
		EUR	ERDF rate	Public	Private	Total	Budget	% of project budget
PP 3 - Cluster Tecnologico Nazionale “Made in Italy”	IT	280,000.00	80.00%	70,000.00	0.00	70,000.00	350,000.00	14.00%
PP 4 - University of Calabria	IT	400,000.00	80.00%	100,000.00	0.00	100,000.00	500,000.00	20.00%
PP 5 - Local Action Group Pollino	IT	240,000.00	80.00%	0.00	60,000.00	60,000.00	300,000.00	12.00%
PP 6 - Calabria Region	IT	340,000.00	80.00%	85,000.00	0.00	85,000.00	425,000.00	17.00%
Total (€)		**2,000,000.00**	**80%**	**440,000.00**	**60,000.00**	**500,000.00**	**2,500,000.00**	**100.00%**

A.3 Additional Information

How did you learn about this call? European Urban Initiative website

Part B - Partnership

B.1 Partnership - summary tables

B.1.1 What is the composition of your Partnership?

Twin4forestLife is implemented by a focused city-led partnership combining urban authorities, scientific expertise, regional wildfire-policy capacity, territorial engagement and communication/uptake skills.
Castrovillari (Main Urban Authority) leads the project as it is directly responsible for improving wildfire prevention and preparedness in an urban-forest interface territory connected to the Pollino area. It ensures political ownership, coordination, local data access, pilot implementation and integration of the Digital Twin into municipal planning routines.
Cassano all'Ionio (Associated Urban Authority) brings a complementary risk context: wildfire/interface-fire exposure combined with river corridors, coastal and tourism-sensitive areas, emergency-access needs and an updated Civil Protection Plan. Their joint participation allows the solution to be tested in two complementary municipal contexts, strengthening robustness and transferability.
UNICAL provides the scientific and technical backbone: designing, developing and validating the Digital Twin, integrating climate, vegetation, drone/sensor, risk, access and logistics data.
Calabria Region ensures alignment with the regional AIB framework, wildfire-risk data, monitoring systems, Control Room/Digital Twin ecosystem and policy uptake, avoiding duplication with existing regional structures.
Cluster Made in Italy leads communication, dissemination, exploitation and innovation uptake, translating technical results into usable messages, stakeholder engagement tools and replication outputs.
GAL Pollino provides territorial knowledge, access to local communities, rural actors and environmental stakeholders, supporting participatory activities and long-term uptake in the Castrovillari/Pollino context.
The partnership is deliberately compact: each partner has a clear operational role and a direct contribution to implementation, sustainability and replication.

B 1.2 How will each Partner contribute to the implementation of the innovative solution?

Castrovillari (MUA) will coordinate the project and ensure political, administrative and operational ownership. It will lead the Castrovillari living lab, provide local datasets, identify priority urban-forest interface areas, validate Digital Twin outputs with municipal services and embed results in local prevention and preparedness practices.
Cassano all'Ionio (AUA) will implement the second living lab. It will provide its updated Civil Protection Plan, wildfire/interface-fire scenarios, emergency areas, COC-related information, access routes, vulnerable assets and communication procedures, testing how the Digital Twin can support preparedness, emergency-access planning and targeted communication.
UNICAL will lead scientific and technical development: designing the data architecture, integrating climate, vegetation, risk, drone/sensor, access, evacuation and water-logistics layers, developing scenario-based decision-support functions and producing technical validation outputs.
Calabria Region will ensure alignment with the regional AIB framework, wildfire-risk data, monitoring systems and Digital Twin ecosystem. It will support access to regional datasets, contribute to validation of use cases and ensure pilots remain complementary to existing regional structures.
Cluster Made in Italy will lead communication, dissemination, exploitation and uptake, translating technical outputs into stakeholder-oriented tools, communication formats and replication materials.
GAL Pollino will support territorial engagement in the Castrovillari/Pollino area, mobilising rural actors, local communities and environmental stakeholders, contributing to participatory validation and long-term uptake.
Responsibilities are distributed to avoid overlaps: municipalities validate and use the solution; UNICAL develops it; Calabria Region ensures regional coherence; Cluster Made in Italy drives uptake; GAL Pollino anchors it locally.

Total number of inhabitants 36,732.00

Role	Number	Country	Name of organisation
Main Urban Authority	**1**	IT	**Comune di Castrovillari**
Associated Urban Authority	**2**	IT	**Comune di Cassano allo Ionio**
Delivery Partner	**3**	IT	**Cluster Tecnologico Nazionale “Made in Italy”**
Delivery Partner	**4**	IT	**Università della Calabria**

Role	Number	Country	Name of organisation
Delivery Partner	5	IT	Pollino Sviluppo S.c.ar.l. — Gruppo di Azione Locale
Delivery Partner	6	IT	Regione Calabria

B.2 - Main Urban Authority

Organisation name (Original)	Comune di Castrovillari
Organisation name (English)	Munipality of Castrovillari
Legal status of the organisation	Public
Organisation type	Local public authority
Member state	Italy
Number of inhabitants	20,409
Comment	UAL Code: IT_078033 UAL Name: Castrovillari Number of inhabitants: 20,439 (ISTAT — source: https://demo.istat.it) Castrovillari is the Main Urban Authority of Twin4forestLife. Located in the Province of Cosenza, Calabria, at the edge of the Pollino National Park, the municipality is directly exposed to wildfire and climate risks at the urban-forest interface. The project will test a municipal Digital Twin to support prevention priorities, vegetation-risk area management, access and water-logistics planning, emergency-area validation and targeted risk communication, integrating regional wildfire intelligence with local municipal data.
Department(s)/unit(s)/division(s) concerned	Municipal Technical Office; Civil Protection Office; Environment and Territorial Planning Department.
NUTS 2	Calabria
Category of region	Less developed

Contact person, legal representative and contact details

Contact Person

Position	Urban Planning Officer
Title	Mr
Forename	Fedele Enrico
Surname	L'Avena
Email	F.lavena@comune.castrovillari.cs.it
Phone Number	39 3893178285

Legal Representative

Position	Major
Title	Ms
Forename	Anna
Surname	De Gaio
Email	protocollo@pec.comune.castrovillari.cs.it
Phone Number	39 09812511

Address

Street	Piazza Municipio 1
Postal Code	87012
Town	Castrovillari

VAT and Cost Option(s)

VAT number	00430920785

VAT recoverable	No
Project Partner Cost Option	Hourly rate for Staff, flat rate for Office & Administration, Travel & Accommodation; real cost for the rest of costs

Total Partner Budget

ERDF (€)	**420,000.00**	**Total (€)**	525,000.00

B.3 - Associated Urban Authority (Partner 2)

Organisation name (Original)	Comune di Cassano allo Ionio
Organisation name (English)	Municipality of Cassano allo Ionio
Legal status of the organisation	Public
Organisation type	Local public authority
Member state	Italy
Number of inhabitants	16,323
Comment	UAL Code: IT_078029 UAL Name: Cassano all'Ionio Number of inhabitants: 18,355 (istituzionale del Comune, dati ISTAT ultimo censimento — https://demo.istat.it) Cassano all'Ionio is the Associated Urban Authority of Twin4forestLife. Located in the Province of Cosenza, Calabria, the municipality combines wildfire and interface-fire exposure with a complex territorial system including river corridors, coastal and tourism-sensitive areas, emergency-access routes and vulnerable assets. Its updated Civil Protection Plan already identifies wildfire/interface-fire scenarios, emergency areas and communication procedures. The project will test the municipal Digital Twin to operationalise these planning layers and support preparedness, emergency-access planning and targeted risk communication.
Department(s)/unit(s)/division(s) concerned	Civil Protection Office; General Affairs Department; Environment and Territorial Planning Office.
NUTS 2	Calabria
Category of region	Less developed

Contact person, legal representative and contact details

Contact Person

Position	Urban Planning Officer	
Title	Mr	
Forename	Mario	
Surname	Oriolo	
Email	oriolomario@gmail.com	
Phone Number	39	3388532956

Legal Representative

Position	Major	
Title	Mr	
Forename	Gianpaolo	
Surname	Iacobini	
Email	sindaco@comune.cassanoalloionio.cs.it	
Phone Number	39	0981 780201

Address

Street	Via G. Amendola 1
Postal Code	87011
Town	Cassano allo Ionio

VAT and Cost Option(s)

VAT number	88000230784
VAT recoverable	No
Project Partner Cost Option	Hourly rate for Staff, flat rate for Office & Administration, Travel & Accommodation; real cost for the rest of costs

Total Partner Budget

ERDF (€)	**320,000.00**	**Total (€)**	400,000.00

B.4 - Delivery Partner (Partner 3)

Organisation name (Original)	Cluster Tecnologico Nazionale “Made in Italy”
Organisation name (English)	Cluster Tecnologico Nazionale “Made in Italy”
Legal status of the organisation	Private
Organisation type	Sectoral agency
Member state	Italy
NUTS 2	Lombardia
Category of region	More developed
Partner description and department/unit/division concerned	Cluster Made in Italy, MINT, is a non-profit association under Italian law and the National Technological Cluster for Design, creativity and Made in Italy. Its statutory mission is to promote and facilitate pre-competitive research in the Italian and European research area, with a focus on Design, creativity and Made in Italy. The Cluster coordinates a public-private partnership aimed at strengthening scientific and entrepreneurial excellence, supporting SMEs in the relevant value chains, fostering synergies between industry, research organisations and public administration, transferring best practices and technologies, promoting specialised training and reinforcing national and international cooperation networks. The Cluster may organise training courses, seminars, events, conferences and workshops; collaborate with universities, research, cultural, public and private institutions; support study, research and information activities; and facilitate networking among members and other actors involved in its initiatives. Website: https://www.clusterminit.it/?lang=en

Contact person, legal representative and contact details

Contact Person

Position	Cluster Manager	
Title	Ms	
Forename	Daniela	
Surname	Fontana	
Email	d.fontana@iuav.it	
Phone Number	39	389 587 9132

Legal Representative

Position	President	
Title	Mr	
Forename	Attilio Alberto	
Surname	Bassi	
Email	bassi@iuav.it	
Phone Number	39	335 681 0607

Address

Street	Piazza San Marco 1
Postal Code	20100
Town	Milan

VAT and Cost Option(s)

VAT number	97799060153
VAT recoverable	No

Project Partner Cost Option	Hourly rate for Staff, flat rate for Office & Administration, Travel & Accommodation; real cost for the rest of costs

Total Partner Budget

ERDF (€)	**280,000.00**	**Total (€)**	350,000.00

B.4 - Delivery Partner (Partner 4)

Organisation name (Original)	Università della Calabria
Organisation name (English)	University of Calabria
Legal status of the organisation	Public
Organisation type	Higher education and research organisations
Member state	Italy
NUTS 2	Calabria
Category of region	Less developed
Partner description and department/unit/division concerned	Description (per il form): The University of Calabria (UNICAL) is a public research university founded in 1968, located in Rende (Cosenza). It provides the scientific and technical backbone of Twin4forestLife through Lab LMSV, responsible for the multi-factor wildfire risk analysis, Digital Twin design and development, data integration and methodological validation. Department concerned: Lab LMSV

Contact person, legal representative and contact details

Contact Person

Position	Full Professor	
Title	Ms	
Forename	Maria	
Surname	Mazzuca	
Email	maria.mazzuca@unical.it	
Phone Number	39	346 416 3276

Legal Representative

Position	Rector	
Title	Mr	
Forename	Gianluigi	
Surname	Greco	
Email	gianluigi.greco@unical.it	
Phone Number	39	0984 4911

Address

Street	Ponte Pietro Bucci
Postal Code	87036
Town	Rende

VAT and Cost Option(s)

VAT number	00419160783
VAT recoverable	No
Project Partner Cost Option	Hourly rate for Staff, flat rate for Office & Administration, Travel & Accommodation; real cost for the rest of costs

Total Partner Budget

ERDF (€)	**400,000.00**	**Total (€)**	500,000.00

B.4 - Delivery Partner (Partner 5)

Organisation name (Original)	Pollino Sviluppo S.c.ar.l. — Gruppo di Azione Locale
Organisation name (English)	Local Action Group Pollino
Legal status of the organisation	Private
Organisation type	Business support organisation
Member state	Italy
NUTS 2	Calabria
Category of region	Less developed
Partner description and department/unit/division concerned	GAL Pollino Sviluppo is a Local Action Group operating in the northern province of Cosenza, covering 17 municipalities, 2 mountain communities, the Pollino National Park and 68 economic operators and associations. It supports local development through LEADER/EAFRD programmes and provides territorial knowledge, community engagement and access to rural, environmental and economic stakeholders in the Castrovillari/Pollino area. In Twin4forestLife, GAL Pollino supports participatory activities, local validation and long-term territorial uptake.

Contact person, legal representative and contact details

Contact Person

Position	Direttore della Società Pollino Sviluppo	
Title	Mr	
Forename	Francesco	
Surname	Arcidiacono	
Email	info@commercialistaarcidiacono.com	
Phone Number	39	328 288 2826

Legal Representative

Position	President	
Title	Mr	
Forename	Domenico	
Surname	Pappaterra	
Email	segreteria@galpollino.it	
Phone Number	39	0981 26919

Address

Street	Via Roma 8
Postal Code	87011
Town	Castrovillari

VAT and Cost Option(s)

VAT number	02154190785
VAT recoverable	No
Project Partner Cost Option	Hourly rate for Staff, flat rate for Office & Administration, Travel & Accommodation; real cost for the rest of costs

Total Partner Budget

ERDF (€)	**240,000.00**	**Total (€)**	300,000.00

B.4 - Delivery Partner (Partner 6)

Organisation name (Original)	Regione Calabria
Organisation name (English)	Calabria Region
Legal status of the organisation	Public
Organisation type	Regional public authority
Member state	Italy
NUTS 2	Calabria
Category of region	Less developed
Partner description and department/unit/division concerned	Calabria Region is the regional public authority responsible for wildfire prevention, forest management and civil protection in Calabria. It manages the regional AIB (Antincendio Boschivo) framework, drone-based monitoring, the regional Control Room and the emerging regional Digital Twin ecosystem for wildfire risk. In Twin4forestLife, Calabria Region acts as Delivery Partner ensuring alignment between the municipal Digital Twin pilots and regional wildfire intelligence, data systems and policy priorities. It supports access to relevant regional datasets, contributes to validation of prevention and preparedness use cases, and ensures that the solutions tested in Castrovillari and Cassano all'Ionio remain interoperable with — and not duplicative of — existing regional AIB, monitoring and emergency-management structures.

Contact person, legal representative and contact details

Contact Person

Position	Dirigente del Settore "Difesa del Suolo, Tutela del Territorio, Prevenzione Calamità"	
Title	Mr	
Forename	Raffaele	
Surname	Mangiardi	
Email	raffaele.mangiardi@regione.calabria.it	
Phone Number	39	0961853419

Legal Representative

Position	President	
Title	Mr	
Forename	Occhiuto	
Surname	Roberto	
Email	presidente@regione.calabria.it	
Phone Number	39	0961858755

Address

Street	Località Germaneto, Viale Europa
Postal Code	88100
Town	Catanzaro

VAT and Cost Option(s)

VAT number	02205340793
VAT recoverable	No
Project Partner Cost Option	Hourly rate for Staff, flat rate for Office & Administration, Travel & Accommodation; real cost for the rest of

costs

Total Partner Budget

ERDF (€)	**340,000.00**	**Total (€)**	425,000.00

Part C - Project description

C.1 Project relevance and innovativeness

C.1.1 What is the main challenge to be addressed by your project?

Twin4forestLife addresses the growing difficulty of small and medium-sized municipalities in managing wildfire risk at the urban-forest, urban-rural and climate-risk interface. In Calabria, wildfire risk is structural: between 2004 and 2020, 15,645 forest fires were recorded, affecting 206,013 hectares, including 133,857 hectares of wooded land. The challenge is not limited to forests; it also concerns peri-urban vegetation, agricultural land, river corridors, access routes, emergency areas and exposed communities.The project focuses on Castrovillari and Cassano all'Ionio, two Calabrian municipalities facing complementary dimensions of the same challenge. Castrovillari (11,837 ha, elevation range 0–2,262 m) is exposed to wildfire and climate risks linked to its urban-forest interface and its position as gateway to the Pollino National Park, where over 28% of the municipal area falls within Natura 2000 sites. The simulated wildfire risk analysis shows that 9.4% of its territory is in high or very high risk classes, concentrated at mid-elevation transition zones where slope, aspect, shrub cover and historical ignition patterns converge. Cassano all'Ionio (14,017 ha, predominantly flat) records a higher mean risk value (2.94 vs 2.47), with 13% of territory in high or very high risk classes, driven by widespread Mediterranean scrub and crops under strong summer water stress, combined with the historical recurrence of ignitions along the agricultural-coastal strip.The challenge is environmental, social and economic. Satellite data (NASA FIRMS VIIRS, 2012–2025) record over 1,150 thermal anomaly points across the two municipalities, with a peak of over 200 events in 2017. 106 burned area perimeters are mapped by EFFIS/Copernicus. Despite this documented risk, existing wildfire-risk maps, civil-protection plans, regional AIB data, emergency areas, access routes and communication procedures remain fragmented and are not yet fully usable as real-time municipal decision-support tools. The core problem is a missing operational layer at municipal level: local authorities need practical tools to transform existing maps, data, alerts and civil-protection procedures into decisions on prevention priorities, vegetation-risk areas, access, firefighting water logistics, emergency-area validation and targeted communication.

C.1.2 What is the current way of working in your city in addressing the challenge presented above?

Castrovillari and Cassano all'Ionio currently address wildfire and climate-risk preparedness through ordinary municipal planning, civil-protection and territorial-management tools. This includes municipal civil-protection procedures, emergency areas, local knowledge of access routes and exposed areas, coordination with regional actors, and the use of existing maps and datasets produced by municipal, regional and operational bodies.
This approach is legally grounded but remains mainly document-based, sectoral and reactive. Information on vegetation, wildfire-risk areas, access constraints, firefighting water logistics, emergency areas, vulnerable assets, weather conditions, regional monitoring and communication needs is available in different formats and under different institutional responsibilities. Municipal staff do not yet have a single operational environment where these layers can be combined, visualised and used to support prevention, preparedness and targeted communication.
In Castrovillari, the current approach relies on municipal territorial knowledge, civil-protection functions and existing planning instruments to manage risks linked to the urban-forest interface, peri-urban vegetation, access routes and prevention needs. These elements are not yet integrated into a dynamic municipal tool able to prioritise prevention actions, simulate operational scenarios or connect local decisions with regional wildfire intelligence.
In Cassano all'Ionio, the municipality has an updated Civil Protection Plan with wildfire/interface-fire scenarios, emergency areas, COC-related functions, access routes and communication procedures. This provides a strong planning basis, but the challenge is to move from static planning layers to operational use: validating emergency areas, identifying access and water-logistics constraints, and communicating risk to differentiated target groups. The water resources analysis confirms this gap: the hydrographic network (2,580 segments, 170 watercourses, 26 reservoirs, 129 registered springs) is richly mapped but its summer availability is not yet integrated into operational preparedness decisions.

C.1.3 What is the proposed innovative solution?

Twin4forestLife will implement and test a municipal Digital Twin for wildfire prevention, preparedness and smart risk communication in Castrovillari and Cassano all'Ionio. The solution will integrate, in one operational environment: regional wildfire intelligence and AIB data; municipal civil-protection information; vegetation and land-use layers (Carta Natura ISPRA/EUNIS, including 2,193 ha of Mediterranean scrub and 1,754 ha of deciduous woodland in the study area); simulated wildfire-risk raster data; climate and weather variables; drone/sensor inputs where available; access and evacuation routes; firefighting water-logistics assets; emergency areas; vulnerable assets; visitor flows; and target-group communication needs.
The Digital Twin will be tested through two complementary living labs. In Castrovillari, it will support prevention-oriented management of the urban-forest interface, helping the municipality identify priority vegetation-risk areas, access constraints and prevention actions in a morphologically complex territory (18.5% of the area with slopes above 60°, 28% within Natura 2000). In Cassano all'Ionio, it will support wildfire/interface-fire preparedness by connecting the Civil Protection Plan with emergency areas, COC functions, access routes, exposed assets and targeted communication procedures in a predominantly flat, high-risk agricultural and coastal landscape.
The innovation lies in transforming fragmented plans, maps and datasets into a dynamic municipal decision-support layer. This is not a new static risk map or a parallel emergency system. It makes regional wildfire intelligence actionable at local level, enabling municipalities to anticipate risk,

prioritise preventive measures, test scenarios, validate emergency areas and communicate tailored information to residents, vulnerable people, farmers, visitors and local operators. The solution will be built on the existing regional AIB framework and Digital Twin ecosystem, providing the missing municipal operational layer without duplicating regional AIB, Control Room or firefighting structures.

C.1.4 What would you like to achieve by implementing the innovative solution?

By implementing Twin4forestLife, Castrovillari and Cassano all'Ionio aim to permanently improve how municipal authorities prevent, prepare for and communicate wildfire/interface-fire risk, moving from fragmented maps and procedures to an integrated, scenario-based and prevention-oriented way of working.
The expected result is a tested municipal Digital Twin enabling local authorities to combine regional wildfire intelligence with municipal data on vegetation, access routes, firefighting water logistics, emergency areas, vulnerable assets and communication needs. Municipal staff will identify prevention priorities, anticipate critical situations, validate operational choices and communicate targeted information before and during high-risk periods.
If proven successful, the solution will make wildfire preparedness a continuous municipal function rather than a seasonal or reactive activity, improving coordination between municipal services, regional AIB structures, civil-protection actors and communities. The lasting contribution will be threefold: a reusable Digital Twin methodology for small and medium-sized urban authorities; municipal procedures and datasets available after the project; and a transferability pathway for other urban-forest territories. Targets: 90 people trained, 6 validated scenarios, 5 target groups covered by smart risk communication, 10 external organisations engaged in transferability.

C.1.5 Where does your inspiration come from?

Twin4forestLife draws on three complementary sources. First, it builds on Calabria's emerging regional wildfire-prevention strategy — AIB framework, drone-based monitoring, regional Control Room and Digital Twin ecosystem — testing how regional intelligence can become operational at municipal level. Second, it draws on existing local planning assets. The multi-factor wildfire risk analysis by Lab LMSV – Università della Calabria (June 2026) integrates vegetation (Carta Natura ISPRA/EUNIS), morphology (DEM 20 m), water resources, historical ignition dynamics (NASA FIRMS VIIRS 2012–2025, EFFIS/Copernicus) and tourist exposure, demonstrating the value of data integration at municipal level. Cassano's updated Civil Protection Plan provides a concrete basis for testing how wildfire scenarios, COC functions, vulnerable assets and communication procedures can become dynamic preparedness tools. Third, it draws on European and international practices on digital twins, climate-risk governance and living-lab approaches, adapting them to small and medium-sized urban authorities. Twin4forestLife combines regional innovation, municipal planning needs and EU-level urban innovation methods to bridge the gap between advanced regional monitoring and everyday local decisions.

C.1.6 To which extent is the project city-led and owned by the Urban Authority(ies)?

Twin4forestLife is fully city-led and owned by the participating Urban Authorities. The project responds to a core public responsibility of Castrovillari and Cassano all'Ionio: protecting residents, vulnerable people, operators, visitors and natural landscapes from wildfire and climate-related risks. Castrovillari, as Main Urban Authority, leads the project politically, administratively and operationally. It coordinates the partnership, defines municipal priorities, provides local data and validates the Digital Twin in its urban-forest interface living lab. It is the primary user and owner of the innovation, ensuring that results are tested against real municipal prevention and preparedness needs. Cassano all'Ionio, as Associated Urban Authority, directly experiments the solution through its living lab, providing civil-protection planning layers, emergency areas, access routes, wildfire scenarios and communication procedures. The partnership was built around the needs of the Urban Authorities. UNICAL, Calabria Region, Cluster Made in Italy and GAL Pollino support the municipalities with technical, regional-policy, communication and territorial expertise, while strategic decisions remain anchored in municipal needs. The solution contributes to the public interest by improving prevention, preparedness and risk communication, strengthening the permanent capacity of both municipalities to use data and scenarios in daily risk governance.

Part C - Project description

C.2 Partnership and co-creation

C.2.1 What are the target groups that will directly benefit from the project?

Twin4forestLife will directly benefit seven target groups. All figures will be validated in WP1.

Municipal decision-makers and technical staff – approx. 30 people. Elected representatives, technical offices, civil-protection functions, planning staff and communication officers in both municipalities. They will benefit from the Digital Twin MVP, risk layers, scenarios, training and protocols for prevention, access, water logistics and emergency-area decisions.
Local civil-protection actors, volunteers and operational stakeholders – approx. 60 people. Municipal volunteers, local police and emergency-support actors. They will benefit from scenario-based training, preparedness exercises and smart communication procedures.
Residents in wildfire/interface-risk areas – approx. 10,000 people. Residents in exposed urban-forest, peri-urban, rural and interface areas will benefit from improved prevention planning and targeted risk communication before and during high-risk periods.
Vulnerable people and users of sensitive assets – approx. 500 people. Elderly, people with reduced mobility, school users and people near sensitive assets. They will benefit from better identification of exposed assets, emergency-area validation and targeted communication.
Farmers, rural landowners, businesses and tourism/nature operators – approx. 250 people. They will benefit from information on vegetation-risk areas, access constraints and preventive behaviours in a territory hosting 74 tourist and naturalistic points of interest.
Visitors and users of natural, forest and cultural areas – approx. 5,000 people. They will benefit from smart risk-communication tools, safer access information and preparedness messages in high-risk periods.
Other municipalities and institutional stakeholders – approx. 30 representatives. Through the replication toolkit and transferability dialogue, they will benefit from a tested model for other wildfire-exposed urban territories.

C.2.2 Which participation and co-creation mechanisms will be implemented in the project?

Twin4forestLife integrates participation and co-creation as operational tools throughout the project lifecycle, not as stand-alone dissemination activities. The goal is to ensure the Digital Twin reflects real local needs, is understood by its users and can be embedded in everyday prevention and preparedness practices.
Participation will involve four main groups: municipal technical and civil-protection staff; regional and operational stakeholders; residents, vulnerable users and local communities; and rural, environmental, tourism and economic actors. GAL Pollino supports territorial engagement, Cluster Made in Italy designs co-creation formats, while UNICAL and the municipalities translate stakeholder inputs into Digital Twin requirements, scenarios and validation criteria.
In WP1, stakeholder mapping and local needs workshops will identify priority wildfire/interface-risk areas, data layers, communication gaps, exposed assets and user requirements, shaping the Digital Twin architecture from the outset.
In WP2, co-design and testing sessions will be organised in both living labs. In Castrovillari, stakeholders will validate urban-forest interface priorities, vegetation-risk areas, access constraints and prevention needs. In Cassano all'Ionio, stakeholders will test how civil-protection layers, emergency areas, access routes and communication procedures can be operationalised through the Digital Twin.
In WP3, preparedness exercises, user feedback sessions and communication testing will assess usability for municipal staff, civil-protection actors, residents, farmers, visitors and local operators. A transferability dialogue with other municipalities will test the replicability of the model.
This approach reflects the New Leipzig Charter principle of shared responsibility: public authorities lead, but citizens and local actors contribute to defining, testing and improving the solution, reducing implementation risks and supporting long-term uptake.

Part C - Project description

C.3 Project objectives, results and outputs

C.3.1 What is your overall objective?

To test a municipal Digital Twin that strengthens wildfire prevention, preparedness and smart risk communication in Castrovillari and Cassano all'Ionio, making regional wildfire intelligence actionable for local decision-making.

C.3.2 What are the three specific objectives supporting the achievement of your overall objective?

Specific objectives
Establish the baseline, governance model and data architecture needed to build a municipal Digital Twin integrating wildfire-risk, climate, civil-protection, access, water-logistics and communication layers.
Deploy and test the municipal Digital Twin in Castrovillari and Cassano all'Ionio to support prevention priorities, wildfire/interface scenarios, access planning, water logistics and emergency-area validation.
Strengthen municipal preparedness, stakeholder uptake and transferability through training, smart risk communication, operational protocols, replication tools and policy alignment with regional systems.

C.3.3 What are the expected results you want to achieve at the end of project implementation? What is the "desired change" you seek in your local context?

Expected results
A shared municipal baseline, governance model and data architecture are established, enabling Castrovillari and Cassano all'Ionio to organise wildfire, climate, civil-protection and communication data for Digital Twin use.
The municipal Digital Twin is tested in two living labs and used by the Urban Authorities to support prevention priorities, wildfire/interface scenarios, access planning, water logistics and emergency-area validation.
Municipal preparedness and uptake are strengthened through trained users, smart risk-communication procedures, operational protocols and a transferability model for other wildfire-exposed urban territories.

C.3.4 Which indicators will support you to measure the expected results described in C.3.3?

Twin4forestLife uses project-specific result indicators measuring municipal capacity, operational preparedness and local usability of wildfire intelligence. All indicators are linked to the three expected results and measured through baseline assessment, Digital Twin testing records, training records, user surveys and validation reports.
RI1 – Integrated data categories (target: 8). Number of wildfire-risk, climate, civil-protection, access, water-logistics, emergency-area and communication data categories integrated into the Digital Twin architecture.
RI2 – Validated municipal use cases (target: 2). Number of Digital Twin use cases formally specified and validated by Castrovillari and Cassano all'Ionio, measured through use-case fiches and approval records.
RI3 – Digital Twin pilot modules tested (target: 2). Number of operational modules tested in real pilot conditions in the two living labs, measured through testing reports and user feedback.
RI4 – Validated wildfire-preparedness scenarios (target: 6). Number of scenarios generated and validated through the Digital Twin, covering prevention priorities, access routes, water logistics and emergency-area validation.
RI5 – Users reporting improved decision capacity (target: 70%). Percentage of trained municipal users reporting improved capacity to use data and scenarios, measured through pre/post-training questionnaires.
RI6 – People trained or involved in preparedness (target: 90). Municipal staff, civil-protection actors and stakeholders trained or involved in

preparedness exercises and Digital Twin testing.
RI7 – Target groups covered by risk communication (target: 5). Number of target groups covered by tested smart risk-communication procedures.
RI8 – Organisations engaged in transferability (target: 10). External municipalities and institutional actors engaged in the transferability pathway through replication workshops and dissemination activities.

C.3.5 Result indicators

Name	Description	Unit	Baseline	Target Value
Population benefiting from protection measures against non-climate related natural risks and risks related to human activities	No structured municipal Digital Twin or integrated wildfire-protection tool currently operational in Castrovillari or Cassano all'Ionio.	percentage of the local population protected	0	15,500
Population benefiting from protection measures against climate related natural disasters (other than floods and forest fires)	No integrated climate-risk decision-support system currently available at municipal level in the two project territories.	percentage of the local population protected	0	15,500
Users of new and upgraded digital services, products and processes	No municipal Digital Twin service currently in use by staff, civil-protection actors or local stakeholders in either municipality.	users/year	0	90

C.3.6 Outputs

Output Number	Project output	Output indicator	Unit	Target value of project expected output(s)
O 1 . 1	Municipal wildfire and forest-risk baseline	Number of cultural and tourism sites supported	cultural and tourism sites	1.00
O 1 . 2	Digital Twin governance and functional architecture	New products and services created	new products/services	1.00
O 1 . 3	Initial communication and stakeholder engagement framework	Participations in joint actions promoting gender equality, equal opportunities and social inclusion	participations	30.00
O 2 . 1	Operational Digital Twin MVP	New products and services created	new products/services	1.00
O 2 . 2	Two tested municipal pilot modules	New products and services created	new products/services	2.00
O 2 . 3	Fire-risk, access and water-logistics scenario package	New products and services created	new products/services	1.00
O 2 . 4	Pilot communication and user engagement package	Stakeholders involved in the preparation and co-implementation of the project	participations of stakeholders	40.00
O 3 . 1	Smart risk communication and preparedness protocol	New products and services created	new products/services	1.00

Output Number	Project output	Output indicator	Unit	Target value of project expected output(s)
O 3 . 2	Trained municipal and local operator network	People supported (trained, upskilled, accompanied or assisted)	persons	90.00
O 3 . 3	Replication toolkit for wildfire-resilient municipalities	New products and services created	new products/services	1.00
O 3 . 4	Impact validation and policy uptake package	New products and services created	new products/services	1.00

C.3.6 Relevant specific objectives of the EU Cohesion policy 2021-2027

Specific objective 2.4 “Promoting climate change adaptation, and disaster risk prevention, and resilience, taking into account eco-system based approaches” (Greener Europe)

Part C - Project description

C.4 Project sustainability scaling up and transferability

C.4.1 How does your project align with existing priorities?

Twin4forestLife is aligned with local, regional and EU priorities on public safety, climate adaptation, environmental protection and data-driven territorial governance.
At local level, the project supports Castrovillari's strategic orientation as gateway to the Pollino area, where environmental protection, tourism and inter-municipal cooperation are connected priorities. The Digital Twin will operationalise priorities on urban-forest interface management, vegetation-risk areas, biodiversity, undergrowth management and prevention of environmental degradation.
In Cassano all'Ionio, the project is aligned with the municipal Civil Protection Plan, which already includes wildfire/interface-fire scenarios, emergency areas, access routes and communication procedures. Twin4forestLife makes these layers operational through scenario-based decision support and smart risk communication.
At regional level, the project aligns with Calabria's AIB framework, wildfire-prevention policy, drone-based monitoring, Control Room and Digital Twin ecosystem. It will not duplicate regional systems; it will create the missing municipal operational layer to use regional wildfire intelligence in local prevention and preparedness.
This alignment strengthens sustainability: the solution will be embedded in municipal planning routines, civil-protection preparedness, regional policy uptake and a replication pathway for other wildfire-exposed municipalities.

C.4.2 How do you intend to sustain the proposed solution after the project ends?

After project end, the municipal Digital Twin and its outputs will be owned by the Urban Authorities. Castrovillari will ensure overall continuity; Cassano all'Ionio will maintain the solution for its local preparedness needs. Both municipalities will integrate tested outputs into their ordinary prevention, civil-protection, territorial-management and communication routines.
Maintenance will follow a light and realistic model. The Urban Authorities will retain municipal datasets, scenarios, protocols and communication tools. UNICAL will define technical maintenance requirements and a handover package. Calabria Region will support alignment with regional AIB data and monitoring systems. Post-project updates will be planned according to municipal resources and future funding opportunities.
Sustainability is not based on keeping a website online, but on embedding a new way of working in municipal governance. Twin4forestLife will leave the Urban Authorities with reusable data, trained staff, tested procedures, validated scenarios and a replicable model — continuing to generate value for residents, vulnerable people, farmers, visitors and other wildfire-exposed municipalities.

C.4.3 Do you see a potential for scaling up the proposed solution?

Yes. Twin4forestLife has strong scale-up potential as a modular municipal model. If proven successful, it can scale in three directions.
First, within the participating municipalities, extending to additional wildfire/interface-risk areas, peri-urban vegetation zones, rural settlements, river corridors and sensitive assets not fully covered during the initial testing phase.
Second, replication by other Calabrian municipalities exposed to wildfire and climate risks. Because the project aligns with Calabria's regional AIB framework and Digital Twin ecosystem, scale-up will not require a parallel regional infrastructure — only adaptation of the tested municipal operational layer.
Third, transfer to other small and medium-sized European urban authorities facing similar risks at the interface between settlements, forests, agricultural land and tourism-sensitive landscapes.
A scale-up pathway will be developed in WP3 through the replication toolkit, transferability dialogue and policy uptake package, defining minimum data requirements, governance model, technical specifications, stakeholder-engagement method and cost/maintenance conditions for adoption elsewhere.

Part D - Work Plan

D.1 Management

D 1.1 How will you coordinate your project?

Twin4forestLife will be coordinated by Castrovillari as Main Urban Authority, responsible for overall project management, administrative coordination, partner communication, progress monitoring, reporting, budget supervision and liaison with EUI authorities.A Project Steering Committee, chaired by Castrovillari and composed of all partners, will meet regularly to take strategic decisions, monitor risks, approve key outputs and ensure alignment between technical work, municipal needs and regional policy. Each Work Package will have one responsible lead for day-to-day coordination of activities, deadlines and deliverables.A Technical and Innovation Group, led by UNICAL with the Urban Authorities and Calabria Region, will coordinate Digital Twin architecture, data integration, pilot testing and technical validation. A Preparedness, Communication and Uptake Group, involving all partners, will coordinate stakeholder engagement, training, risk communication and transferability.Internal communication will use monthly coordination meetings, WP working sessions, shared document repository, deliverable calendar and risk/action tracker. This structure ensures city-led coordination while linking management, Digital Twin development, living-lab implementation, communication and sustainability in one integrated governance system.

D 1.2 Which measures will you take to ensure quality in your project?

Quality will be ensured through a structured Quality Assurance and Evaluation Framework coordinated by Castrovillari, with technical support from UNICAL and contributions from all partners.
Four measures will be applied. First, each Work Package will have a responsible lead, activity calendar, output specifications, internal deadlines and validation steps. Second, all key outputs will undergo partner review: UNICAL checks technical outputs; the Urban Authorities validate municipal usability; Calabria Region verifies regional coherence; Cluster Made in Italy and GAL Pollino review communication outputs. Third, a risk and issue tracker will be reviewed monthly to identify delays, data gaps or usability concerns in time for corrective action. Fourth, all deliverables will be assessed against relevance, feasibility, usability, consistency with the Work Plan and contribution to expected results.
Evaluation will combine baseline assessment, pilot-testing evidence, user feedback, training surveys, scenario-validation records and KPI monitoring. The Steering Committee will use evaluation findings to take corrective decisions during implementation.

D 1.3 What will be the general approach you will follow to communicate about your project?

Communication will be coordinated by Cluster Made in Italy, in cooperation with Castrovillari and all partners. It will be embedded in the Work Packages as an operational function supporting participation, testing, uptake and transferability — not as a separate promotional activity.
Three objectives will guide communication: making the project understandable to non-technical audiences; supporting use of the Digital Twin by municipal staff and stakeholders; and transferring results to other wildfire-exposed territories. Messages, channels and formats will be adapted to different audiences: residents, vulnerable people, farmers, local operators, visitors, civil-protection actors, regional stakeholders and other municipalities.
Cluster Made in Italy will prepare the communication framework, visual identity and dissemination materials. The municipalities will ensure communication reflects local needs. UNICAL will translate technical results into accessible content. Calabria Region will support policy alignment. GAL Pollino will facilitate community outreach. All communication will comply with EUI visibility requirements and focus on practical uptake, preparedness and institutional learning.

D 1.4 How do you foresee the financial management of the project and reporting procedures for activities and budget (within the partnership and towards the programme)?

Financial management will be coordinated by Castrovillari as MUA, responsible for overall budget monitoring, reporting to the programme and payment flows to partners according to EUI rules and the Subsidy Contract.
Each partner will appoint a financial contact responsible for internal budget monitoring, expenditure eligibility verification and timely submission of financial information to the MUA. Partner budgets will be monitored by Work Package, cost category and activity, ensuring expenditure remains consistent with the approved Work Plan.
Internal financial reporting will be aligned with project monitoring meetings and programme reporting deadlines. Partners will submit expenditure updates, activity evidence and budget forecasts using common templates. The MUA will consolidate inputs, check consistency between financial and activity progress, and identify deviations requiring corrective action.
No partner will shift resources or subcontract activities without prior internal and, where required, programme approval. The Steering Committee will regularly review budget absorption, delays and procurement risks, ensuring resources remain proportionate, traceable and connected to project delivery.

Part D - Work plan

WP Nr.	Title	Start date	End date	Work Package Budget
1	Baseline, governance and Digital Twin architecture	01/03/2027	31/08/2028	375,000.00
2	Digital Twin deployment and wildfire-risk pilot actions	31/07/2027	31/08/2028	1,750,000.00
3	Operational preparedness, uptake and replication	30/09/2027	28/02/2029	327,500.00

Work Plan Per Work Packages - Work Package 1 Thematic

Title Baseline, governance and Digital Twin architecture

Start Date 01/03/2027

End Date 31/08/2028

Partners Involvement

Responsible Partner PP 1 - Munipality of Castrovillari

Involved Partners PP 2 - Municipality of Cassano allo Ionio; PP 3 - Cluster Tecnologico Nazionale "Made in Italy"; PP 4 - University of Calabria; PP 5 - Local Action Group Pollino; PP 6 - Calabria Region

Project Specific Objective Establish the baseline, governance model and data architecture needed to build a municipal Digital Twin integrating wildfire-risk, climate, civil-protection, access, water-logistics and communication layers.

Summary

WP1 sets up the strategic, technical and operational foundations of Twin4forestLife. It will establish the shared baseline, governance model and Digital Twin architecture required to transform fragmented wildfire, climate, civil-protection and territorial data into a usable municipal decision-support environment.
The Municipality of Castrovillari, as Main Urban Authority, will coordinate the WP and ensure city ownership. Cassano all'Ionio will provide its civil-protection planning layers, wildfire/interface-fire scenarios, emergency areas, access routes and local procedures. UNICAL will lead the technical work on data architecture, interoperability, functional requirements and Digital Twin use cases. Calabria Region will ensure alignment with the regional AIB framework, wildfire-risk data, monitoring systems and regional Digital Twin ecosystem. Cluster Made in Italy and GAL Pollino will support stakeholder mapping, communication needs analysis and participatory engagement.
WP1 will produce three main outputs: a municipal wildfire and forest-risk baseline, a Digital Twin governance and functional architecture, and an initial communication and stakeholder engagement framework. These outputs will directly feed WP2, where the Digital Twin will be deployed and tested, and WP3, where preparedness, communication and replication activities will be implemented.

Communication Objective of this Work Package

The communication objective of WP1 is to establish a shared understanding of the project among partners, municipal services, regional actors and local stakeholders. Communication will support stakeholder identification, needs collection, user-requirement definition and early engagement in the two living labs. It will ensure that the Digital Twin is not perceived as a purely technical tool, but as a municipal prevention and preparedness solution shaped by users, communities and operational needs.

Activities and Deliverables

A 1.1 - Project governance, coordination setup and quality framework

Title Project governance, coordination setup and quality framework

Start date 01/03/2027

End date 30/04/2027

Activity description and Partners involved

This activity will establish the project governance system, internal coordination procedures, quality assurance process, reporting flows, risk tracker and decision-making structure. Castrovillari will lead as Main Urban Authority. All partners will appoint operational and financial contacts. UNICAL will contribute to technical coordination procedures, Calabria Region to policy alignment, Cluster Made in Italy to communication coordination, and GAL Pollino to territorial engagement arrangements.

D 1.1.1 - Project governance and quality plan

Deliverable title	Target value	Delivery date
Project governance and quality plan	1	30/04/2027

Description

Governance, reporting, risk, quality and coordination procedures approved by all partners. Responsible: Castrovillari, with inputs from all partners.

D 1.1.2 - Partner roles and coordination matrix

Deliverable title	Target value	Delivery date
Partner roles and coordination matrix	1	30/04/2027

Description

Matrix defining partner responsibilities, WP/activity roles, contacts, internal deadlines and validation steps. Responsible: Castrovillari.

A 1.2 - Municipal wildfire, climate and civil-protection baseline

Title	Start date	End date
Municipal wildfire, climate and civil-protection baseline	01/03/2027	30/06/2027

Activity description and Partners involved

This activity will collect and organise the baseline information needed for the Digital Twin. It will cover wildfire-risk areas, vegetation and land-use layers, climate-related variables, civil-protection planning information, emergency areas, access and evacuation routes, firefighting water-logistics assets, vulnerable assets and communication needs. Castrovillari and Cassano will provide municipal data. UNICAL will structure the baseline methodology. Calabria Region will support alignment with regional AIB and wildfire-risk information. GAL Pollino will contribute territorial knowledge.

D 1.2.1 - Municipal data inventory

Deliverable title	Target value	Delivery date
Municipal data inventory	1	31/05/2027

Description

Inventory of available municipal, regional and territorial datasets relevant to wildfire prevention and preparedness. Responsible: UNICAL with Castrovillari, Cassano and Calabria Region.

D 1.2.2 - Baseline risk and asset mapping report

Deliverable title	Target value	Delivery date
Baseline risk and asset mapping report	1	30/06/2027

Description

Report covering priority risk areas, exposed assets, access routes, water-logistics elements and data gaps in both municipalities. Responsible: UNICAL; validation by Urban Authorities.

A 1.3 - Digital Twin requirements, data governance and functional architecture

Title: Digital Twin requirements, data governance and functional architecture
Start date: 31/05/2027
End date: 31/08/2027

Activity description and Partners involved

This activity will define the Digital Twin functional requirements, data-governance model, interoperability approach and municipal use cases. UNICAL will lead the technical design. Castrovillari and Cassano will validate user needs and operational priorities. Calabria Region will verify consistency with regional AIB systems, monitoring tools and Digital Twin ecosystem. The activity will define what the Digital Twin must do, which data layers it will integrate, and how it will support municipal prevention and preparedness decisions.

D 1.3.1 - Digital Twin functional requirements

Deliverable title: Digital Twin functional requirements
Target value: 1
Delivery date: 31/07/2027

Description

Functional requirements and user needs for the municipal Digital Twin, including prevention, preparedness, access, water-logistics and communication functions. Responsible: UNICAL.

D 1.3.2 - Data governance and architecture blueprint

Deliverable title: Data governance and architecture blueprint
Target value: 1
Delivery date: 31/08/2027

Description

Technical blueprint defining data layers, interoperability rules, governance arrangements, access rights and architecture for the Digital Twin MVP. Responsible: UNICAL with Calabria Region.

A 1.4 - Stakeholder mapping, communication needs and co-creation setup

Title: Stakeholder mapping, communication needs and co-creation setup
Start date: 30/04/2027
End date: 31/08/2028

Activity description and Partners involved

This activity will identify stakeholders, target groups and communication needs for both living labs. Cluster Made in Italy will lead the communication and engagement framework. GAL Pollino will support local outreach, especially in the Castrovillari/Pollino context. Castrovillari and Cassano will identify municipal services, civil-protection actors, residents, vulnerable groups, rural operators, tourism actors and community stakeholders. UNICAL will translate stakeholder inputs into Digital Twin user requirements.

D 1.4.1 - Stakeholder and target-group map

Deliverable title: Stakeholder and target-group map
Target value: 1
Delivery date: 30/06/2027

Description

Map of key target groups and stakeholders, including municipal staff, civil-protection actors, residents, vulnerable users, farmers, visitors and local operators. Responsible: Cluster Made in Italy and GAL Pollino.

D 1.4.2 - Communication and co-creation framework

Deliverable title: Communication and co-creation framework
Target value: 1
Delivery date: 31/08/2027

Description
Framework defining messages, channels, participation formats and communication needs for WP2 testing and WP3 uptake. Responsible: Cluster Made in Italy.

Outputs

O.1.1 - Municipal wildfire and forest-risk baseline

Title: Municipal wildfire and forest-risk baseline
Target value: 1
Delivery date: 30/06/2027

Output indicator: Number of cultural and tourism sites supported
Unit: cultural and tourism sites

Description
Baseline package covering wildfire/interface-risk areas, vegetation and land-use layers, emergency areas, access routes, water-logistics elements, exposed assets and data gaps for Castrovillari and Cassano all'Ionio. Responsible: UNICAL with Urban Authorities and Calabria Region.

O.1.2 - Digital Twin governance and functional architecture

Title: Digital Twin governance and functional architecture
Target value: 1
Delivery date: 31/08/2027

Output indicator: New products and services created
Unit: new products/services

Description
Architecture package defining data governance, interoperability, functional requirements, municipal use cases and technical specifications for the Digital Twin MVP. Responsible: UNICAL, validated by Castrovillari, Cassano and Calabria Region.

O.1.3 - Initial communication and stakeholder engagement framework

Title	Target value	Delivery date
Initial communication and stakeholder engagement framework	30	31/08/2027

Output indicator	Unit
Participations in joint actions promoting gender equality, equal opportunities and social inclusion	participations

Description

Framework identifying target groups, stakeholders, communication needs, participation methods and co-creation formats for the two living labs and project uptake. Responsible: Cluster Made in Italy with GAL Pollino and Urban Authorities.

Investment(s)

Work Plan Per Work Packages - Work Package 2 Thematic

Title Digital Twin deployment and wildfire-risk pilot actions

Start Date 31/07/2027

End Date 31/08/2028

Partners Involvement

Responsible Partner PP 4 - University of Calabria

Involved Partners PP 1 - Munipality of Castrovillari; PP 2 - Municipality of Cassano allo Ionio; PP 3 - Cluster Tecnologico Nazionale "Made in Italy"; PP 5 - Local Action Group Pollino; PP 6 - Calabria Region

Project Specific Objective Deploy and test the municipal Digital Twin in Castrovillari and Cassano all'Ionio to support prevention priorities, wildfire/interface scenarios, access planning, water logistics and emergency-area validation.

Summary

WP2 is the core experimentation phase of Twin4forestLife. It will deploy and test the municipal Digital Twin developed from the baseline, data architecture and user requirements defined in WP1. The WP will transform wildfire-risk, climate, vegetation, civil-protection, access, water-logistics, emergency-area and communication data into operational municipal use cases.
UNICAL will lead the technical development, integration and testing of the Digital Twin MVP. Castrovillari and Cassano all'Ionio will act as real-life testing environments and validate the solution against their municipal prevention and preparedness needs. Calabria Region will ensure coherence with the regional AIB framework, wildfire-risk data, monitoring systems and regional Digital Twin ecosystem, avoiding duplication with existing regional functions. Cluster Made in Italy and GAL Pollino will support stakeholder engagement, user feedback and communication around the pilot testing. The Digital Twin will be tested through two complementary pilot modules. In Castrovillari, the focus will be the urban-forest interface, vegetation-risk areas, access constraints and prevention priorities. In Cassano all'Ionio, the focus will be wildfire/interface preparedness, emergency areas, COC-related information, access routes, vulnerable assets and smart communication needs. WP2 will deliver the operational Digital Twin MVP, a scenario package on wildfire risk, and a pilot communication and user engagement package.

Communication Objective of this Work Package

The communication objective of WP2 is to support testing, user engagement and practical understanding of the Digital Twin in the two living labs. Communication will translate technical functions into clear messages for municipal staff, civil-protection actors, local stakeholders and communities. It will collect feedback, support demonstrations, explain pilot scenarios and prepare the ground for WP3 preparedness, smart risk communication and transferability activities.

Activities and Deliverables

A 2.1 - Digital Twin MVP development and technical integration

Title Digital Twin MVP development and technical integration

Start date 31/07/2027

End date 31/12/2027

Activity description and Partners involved

UNICAL will develop the Digital Twin MVP based on WP1 requirements and architecture. The activity will integrate selected data layers on wildfire risk, vegetation, climate variables, civil-protection assets, access routes, emergency areas, water logistics and communication needs. Castrovillari and Cassano will provide municipal validation and access to local data. Calabria Region will support coherence with regional AIB and monitoring systems.

D 2.1.1 - Digital Twin MVP prototype

Deliverable title	Target value	Delivery date
Digital Twin MVP prototype	1	31/12/2027

Description

First operational prototype integrating the core data layers and municipal use-case functions. Responsible: UNICAL; validation by Urban Authorities and Calabria Region.

D 2.1.2 - Technical integration report

Deliverable title	Target value	Delivery date
Technical integration report	1	31/12/2027

Description

Report documenting integrated datasets, technical choices, interoperability approach, data gaps and limits of the MVP. Responsible: UNICAL.

A 2.2 - Castrovillari urban-forest interface pilot module

Title	Start date	End date
Castrovillari urban-forest interface pilot module	31/10/2027	31/05/2028

Activity description and Partners involved

This activity will test the Digital Twin in Castrovillari, focusing on urban-forest interface prevention. The pilot will identify priority vegetation-risk areas, access constraints, exposed assets, prevention needs and potential water-logistics elements. Castrovillari will lead local validation and municipal testing. UNICAL will configure the pilot module. Calabria Region will ensure consistency with regional AIB information. GAL Pollino will support territorial engagement.

D 2.2.1 - Castrovillari pilot configuration

Deliverable title	Target value	Delivery date
Castrovillari pilot configuration	1	31/01/2028

Description

Configuration of the Digital Twin module for Castrovillari, including selected local layers, priority areas and testing parameters. Responsible: UNICAL and Castrovillari.

D 2.2.2 - Castrovillari pilot validation report

Deliverable title	Target value	Delivery date
Castrovillari pilot validation report	1	31/05/2028

Description

Report on testing results, user feedback, prevention priorities, access constraints and lessons learned from the Castrovillari living lab. Responsible: Castrovillari with UNICAL.

A 2.3 - Cassano wildfire-interface preparedness pilot module

Title: Cassano wildfire-interface preparedness pilot module
Start date: 31/10/2027
End date: 31/05/2028

Activity description and Partners involved

This activity will test the Digital Twin in Cassano all'Ionio, focusing on wildfire/interface preparedness, emergency areas, COC-related functions, access routes, vulnerable assets and communication needs. Cassano will lead local validation and provide civil-protection planning information. UNICAL will configure the pilot module. Calabria Region will support alignment with regional AIB procedures and data. Cluster Made in Italy will support communication-related testing.

D 2.3.1 - Cassano pilot configuration

Deliverable title: Cassano pilot configuration
Target value: 1
Delivery date: 31/01/2028

Description

Configuration of the Digital Twin module for Cassano, including wildfire/interface layers, emergency areas, access routes and exposed assets. Responsible: UNICAL and Cassano.

D 2.3.2 - Cassano pilot validation report

Deliverable title: Cassano pilot validation report
Target value: 1
Delivery date: 31/05/2028

Description

Report on testing results, user feedback, emergency-area validation, access/logistics issues and preparedness lessons from the Cassano living lab. Responsible: Cassano with UNICAL.

A 2.4 - Scenario modelling for prevention, access and water logistics

Title: Scenario modelling for prevention, access and water logistics
Start date: 31/12/2027
End date: 31/07/2028

Activity description and Partners involved

This activity will use the Digital Twin to generate and validate operational scenarios for wildfire prevention, access constraints, evacuation-support routes, firefighting water logistics and emergency-area suitability. UNICAL will coordinate scenario modelling. Castrovillari and Cassano will validate scenarios with municipal services. Calabria Region will check regional coherence. Outputs will support WP3 preparedness exercises and protocols.

D 2.4.1 - Wildfire-preparedness scenario set

Deliverable title: Wildfire-preparedness scenario set
Target value: 6
Delivery date: 30/06/2028

Description

Set of at least six scenarios covering prevention priorities, access routes, water logistics and emergency-area validation across the two municipalities. Responsible: UNICAL.

D 2.4.2 - Municipal scenario validation record

Deliverable title	Target value	Delivery date
Municipal scenario validation record	1	31/07/2028

Description

Consolidated validation record documenting municipal feedback, scenario usability, operational limits and corrective actions. Responsible: Castrovillari and Cassano with UNICAL.

A 2.5 - Pilot demonstrations, user feedback and technical refinement

Title	Start date	End date
Pilot demonstrations, user feedback and technical refinement	31/03/2028	31/08/2028

Activity description and Partners involved

This activity will organise pilot demonstrations and user feedback sessions to refine the Digital Twin before its transfer to WP3 preparedness and uptake activities. UNICAL will lead technical refinement. Castrovillari and Cassano will involve municipal users and local stakeholders. Cluster Made in Italy will support user-friendly presentation formats and feedback collection. GAL Pollino will support community and territorial participation.

D 2.5.1 - Pilot demonstration sessions

Deliverable title	Target value	Delivery date
Pilot demonstration sessions	4	31/07/2028

Description

At least four demonstration/testing sessions involving municipal staff, civil-protection actors and local stakeholders in the two living labs. Responsible: UNICAL and Urban Authorities.

D 2.5.2 - Digital Twin refinement report

Deliverable title	Target value	Delivery date
Digital Twin refinement report	1	31/08/2028

Description

Report documenting user feedback, technical improvements, usability issues and final refinements to be used in WP3. Responsible: UNICAL with Cluster Made in Italy.

A 2.6 - Pilot communication and stakeholder engagement

Title	Start date	End date
Pilot communication and stakeholder engagement	31/10/2027	31/08/2028

Activity description and Partners involved

This communication-related activity will support the practical understanding and local testing of the Digital Twin. Cluster Made in Italy will coordinate communication formats, pilot factsheets, stakeholder messages and feedback tools. Castrovillari and Cassano will adapt

messages to local target groups. UNICAL will provide technical content in accessible language. GAL Pollino will support territorial outreach, and Calabria Region will ensure consistency with regional AIB communication.

D 2.6.1 - Pilot communication package

Deliverable title: Pilot communication package
Target value: 1
Delivery date: 31/03/2028

Description
User-friendly materials explaining the Digital Twin, pilot scenarios, target groups and testing activities in the two living labs. Responsible: Cluster Made in Italy.

D 2.6.2 - Stakeholder feedback summary

Deliverable title: Stakeholder feedback summary
Target value: 1
Delivery date: 31/08/2028

Description
Summary of feedback collected from municipal users, local actors and stakeholders during pilot testing. Responsible: Cluster Made in Italy with GAL Pollino.

Outputs

O.2.1 - Operational Digital Twin MVP

Title: Operational Digital Twin MVP
Target value: 1
Delivery date: 31/12/2027

Output indicator: New products and services created
Unit: new products/services

Description
Operational municipal Digital Twin MVP integrating selected wildfire-risk, climate, vegetation, civil-protection, access, water-logistics, emergency-area and communication layers. Responsible: UNICAL; validated by Urban Authorities and Calabria Region.

O.2.2 - Two tested municipal pilot modules

Title: Two tested municipal pilot modules
Target value: 2
Delivery date: 31/05/2028

Output indicator: New products and services created
Unit: new products/services

Description
Two Digital Twin pilot modules tested in real municipal conditions: Castrovillari urban-forest interface and Cassano wildfire-interface preparedness. Responsible: UNICAL with Castrovillari and Cassano.

O.2.3 - Fire-risk, access and water-logistics scenario package

Title	Target value	Delivery date
Fire-risk, access and water-logistics scenario package	1	31/07/2028

Output indicator	Unit
New products and services created	new products/services

Description

Validated scenario package covering prevention priorities, wildfire/interface-risk situations, access routes, firefighting water logistics and emergency-area validation. Responsible: UNICAL with Urban Authorities and Calabria Region.

O.2.4 - Pilot communication and user engagement package

Title	Target value	Delivery date
Pilot communication and user engagement package	40	31/08/2028

Output indicator	Unit
Stakeholders involved in the preparation and co-implementation of the project	participations of stakeholders

Description

Communication and engagement package supporting pilot demonstrations, user feedback, stakeholder understanding and preparation for WP3 preparedness and uptake activities. Responsible: Cluster Made in Italy with GAL Pollino.

Investment(s)

I.2.1 - Municipal Digital Twin and smart preparedness deployment package

Title	Budget	Delivery date
Municipal Digital Twin and smart preparedness deployment package	450,000.00	31/08/2028

Description

Twin4forestLife will invest in a municipal Digital Twin deployment package needed to test wildfire prevention, preparedness and smart risk communication in Castrovillari and Cassano all'Ionio. The investment will include: Digital Twin MVP/dashboard development and deployment; GIS/data-integration components; interoperability interfaces with municipal and regional datasets; secure hosting/cloud or server services during the project; two municipal user access/configuration environments; limited field data-capture or monitoring equipment only where essential to fill critical data gaps; and smart communication interfaces such as QR-based information points, templates or digital communication tools for pilot testing.
The investment will not create a parallel AIB, Control Room or firefighting system. It will create the municipal operational layer needed to make existing regional wildfire intelligence, municipal civil-protection data, access routes, emergency areas, water-logistics information and communication procedures usable by local authorities.

Investment 1

Title	Municipal Digital Twin and smart preparedness deployment package
Investment Description	Twin4forestLife will invest in a municipal Digital Twin deployment package needed to test wildfire prevention, preparedness and smart risk communication in Castrovillari and Cassano all'Ionio. The investment will include: Digital Twin

(including indicative budget of the main cost items)	MVP/dashboard development and deployment; GIS/data-integration components; interoperability interfaces with municipal and regional datasets; secure hosting/cloud or server services during the project; two municipal user access/configuration environments; limited field data-capture or monitoring equipment only where essential to fill critical data gaps; and smart communication interfaces such as QR-based information points, templates or digital communication tools for pilot testing. The investment will not create a parallel AIB, Control Room or firefighting system. It will create the municipal operational layer needed to make existing regional wildfire intelligence, municipal civil-protection data, access routes, emergency areas, water-logistics information and communication procedures usable by local authorities.
Delivery date	31/08/2028

Investment aspects	Questions	Project answers
Justification of the investment	Explain why this investment is needed.	The investment is needed because the project cannot achieve its core outputs through studies, meetings or training alone. The main expected change is the operational use of a municipal Digital Twin by Castrovillari and Cassano all'Ionio. This requires a concrete deployment package: dashboard, data-integration environment, interoperability components, pilot configurations, user access, testing interfaces and limited data-capture tools where essential. Without this investment, the project would remain a planning exercise and would not test a real innovative solution in municipal operations.
	Clearly describe the thematic relevance of the investment.	The investment is directly relevant to the call topic Security, safety and preparedness and to the cross-cutting priority Climate action, environment and clean energy. It supports wildfire prevention, climate-risk preparedness, emergency-area validation, access and water-logistics planning and smart risk communication. It addresses the gap identified in the project: existing data, maps and procedures are fragmented and not yet transformed into actionable municipal decision support. The investment makes the innovation tangible and testable.
	Describe who is benefiting (e.g., Partners, city, target groups, etc.) from this investment, and in what way.	The direct beneficiaries are the Urban Authorities of Castrovillari and Cassano all'Ionio, which will use the Digital Twin to support prevention and preparedness decisions. Municipal technical staff, civil-protection functions, local operators and volunteers will benefit from clearer scenarios, risk layers and operational information. Residents, vulnerable people, farmers, visitors and local economic operators will benefit indirectly through better preparedness, targeted communication and safer management of exposed areas. UNICAL and Calabria Region will also benefit from validation evidence useful for technical and policy uptake.
	Please clarify which problem it tackles, which findings you expect from it, how it can be replicated, and how the experience coming from it will be used for the benefit of the programme area.	The investment tackles the municipal gap between regional wildfire intelligence and local operational use. It will test whether small and medium-sized urban authorities can use a Digital Twin to combine wildfire-risk layers, vegetation data, access routes, water-logistics assets, emergency areas and communication needs in one decision-support environment. Expected findings concern data readiness, usability, municipal workflows, minimum technical requirements, maintenance needs and replication conditions. The experience will feed the replication toolkit and policy uptake package, supporting other wildfire-exposed municipalities in Calabria and beyond.
	Describe the location of the physical investment; if possible, a specific	The investment will be deployed and tested in the two municipal pilot areas. One is the Municipality of Castrovillari Municipal offices and selected urban-forest interface areas to be defined during WP1 baseline and WP2 pilot configuration. The secondo is the Municipality of Cassano all'Ionio Municipal offices, COC-

Location of the investment	address where the investment will be located.	related operational context and selected wildfire/interface-risk areas, emergency-access routes and exposed assets to be defined during WP1 baseline and WP2 pilot configuration. The Digital Twin software/dashboard may be hosted through secure cloud/server services managed by the responsible technical partner and made accessible to the Urban Authorities.

Country	NUTS 2 level	NUTS 3 level
Italy	Calabria	Cosenza

Investment documentation	Please list all technical requirements and permissions (e.g., building permits) required for the investment according to the respective national legislation.	The investment will require technical specifications for the Digital Twin MVP, data-governance and interoperability rules, cybersecurity and access-management requirements, GDPR/privacy assessment where personal or mobility-related data are concerned, procurement documentation for software/services/equipment, and inventory/asset registration by the owning partner(s). No building permit is expected, as no construction works are foreseen. Any field equipment or communication device installed in public space will be subject to municipal authorisation, site suitability checks and compliance with applicable procurement, safety and public-space rules.
Ownership	Who owns the site where the investment is located?	The municipal sites and operational premises used for the investment are owned or legally controlled by the relevant Urban Authority: the Municipality of Castrovillari for the Castrovillari pilot environment and the Municipality of Cassano all'Ionio for the Cassano pilot environment. Any temporary field installation or communication point will be placed only in locations where the relevant municipality has ownership, competence or explicit authorisation.
	Who will retain ownership of the investment at the end of the project?	Ownership of the investment will remain within the Project Partnership for at least five years following the final payment to the Main Urban Authority. The Urban Authorities will retain ownership or usage rights over the municipal configurations, datasets, scenarios, protocols and local deployment components. Technical components developed by UNICAL will be transferred or licensed for municipal use according to the Partnership Agreement, ensuring continued access by Castrovillari and Cassano all'Ionio after project end.
	Who will take care of the maintenance of the investment? How will this be done?	Maintenance will be organised through a light post-project model. Castrovillari and Cassano all'Ionio will maintain and update municipal datasets, scenarios, user access and local procedures within their ordinary prevention and civil-protection routines. UNICAL will provide the technical handover package, documentation and maintenance requirements. Calabria Region will support continued alignment with regional AIB data and policy frameworks. Any advanced technical upgrades after the project will depend on municipal resources, regional support or future funding, but the tested outputs will remain usable without creating a parallel regional infrastructure.
Indicative investment total budget	Please provide the total budget required for the investment.	450,000.00
	Please provide an indicative list of costs, including specific items and	- Digital Twin MVP/dashboard development and configuration: 170,000 EUR - GIS/data integration, interoperability interfaces and API configuration: 95,000 EUR - Secure hosting/cloud/server services, cybersecurity and access management during project implementation: 45,000 EUR

Indicative breakdown of the investment costs	estimated amounts, that will be required for the investment. The amounts presented in this field should be equal to the total budget required for the investment (see previous line).	- Municipal pilot configuration for Castrovillari and Cassano all'Ionio: 55,000 EUR - Limited field data-capture / monitoring equipment where essential to fill data gaps: 45,000 EUR - Smart communication interfaces and QR/digital preparedness tools for pilot testing: 25,000 EUR - Testing, technical documentation, handover and maintenance specifications: 15,000 EUR

Work Plan Per Work Packages - Work Package 3 Thematic

Title	Operational preparedness, uptake and replication
Start Date	30/09/2027
End Date	28/02/2029

Partners Involvement

Responsible Partner	PP 4 - University of Calabria
Involved Partners	PP 1 - Munipality of Castrovillari; PP 2 - Municipality of Cassano allo Ionio; PP 3 - Cluster Tecnologico Nazionale "Made in Italy"; PP 5 - Local Action Group Pollino; PP 6 - Calabria Region
Project Specific Objective	Strengthen municipal preparedness, stakeholder uptake and transferability through training, smart risk communication, operational protocols, replication tools and policy alignment with regional systems.

Summary

WP3 transforms the Digital Twin tested in WP2 into practical municipal preparedness, communication and replication tools. Its purpose is to ensure that Twin4forestLife does not remain a technical pilot, but becomes a usable municipal innovation for wildfire prevention, preparedness, risk communication and policy uptake.
UNICAL will coordinate WP3, ensuring methodological continuity between the Digital Twin architecture, pilot testing, scenario validation, training, technical handover and replication model. Castrovillari and Cassano all'Ionio will validate the operational use of the Digital Twin in municipal routines, preparedness exercises and local communication procedures. MINT will lead the translation of technical results into accessible communication formats, smart risk-communication tools, dissemination materials and replication-oriented outputs. Calabria Region will ensure alignment with the regional AIB framework, wildfire-monitoring systems and policy uptake. GAL Pollino will support territorial engagement, community involvement and local dissemination.
WP3 will deliver a smart risk communication and preparedness protocol, training and exercises for municipal and local users, a replication toolkit for wildfire-resilient municipalities, and an impact validation and policy uptake package. The replication dimension will be implemented through a structured transferability with external municipalities and institutional stakeholders.

Communication Objective of this Work Package

The communication objective of WP3 is to support uptake, behavioural preparedness and transferability of Twin4forestLife results. Communication will help municipal users, civil-protection actors, residents, vulnerable people, farmers, visitors and local operators understand how to use risk information and preparedness messages. It will also package the tested model for other municipalities and institutional stakeholders, supporting replication and policy uptake beyond the two living labs.

Activities and Deliverables

A 3.1 - Preparedness protocol and municipal operating procedures

Title	Start date	End date
Preparedness protocol and municipal operating procedures	30/09/2027	30/09/2028

Activity description and Partners involved

This activity will translate Digital Twin scenarios and WP2 pilot findings into municipal preparedness procedures. Castrovillari and Cassano will validate how the Digital Twin can support prevention priorities, emergency-area validation, access and water-logistics planning and targeted communication. Cluster Made in Italy will structure the protocol in accessible operational formats. UNICAL will provide technical inputs, while Calabria Region will verify coherence with the regional AIB framework.

D 3.1.1 - Preparedness protocol draft

Deliverable title	Target value	Delivery date
Preparedness protocol draft	1	30/06/2028

Description

Draft protocol translating Digital Twin scenarios into municipal prevention, access, water-logistics, emergency-area and communication procedures. Responsible: Cluster Made in Italy with Urban Authorities and UNICAL.

D 3.1.2 - Final preparedness protocol

Deliverable title	Target value	Delivery date
Final preparedness protocol	1	30/09/2028

Description

Final operational protocol validated by Castrovillari and Cassano all'Ionio and checked for regional coherence by Calabria Region. Responsible: Cluster Made in Italy.

A 3.2 - Training and preparedness exercises for municipal users

Title	Start date	End date
Training and preparedness exercises for municipal users	31/03/2028	30/11/2028

Activity description and Partners involved

This activity will train municipal staff, civil-protection actors and local operators in the use of the Digital Twin, scenario interpretation and smart risk communication. UNICAL will deliver technical training on Digital Twin functions. Castrovillari and Cassano will mobilise municipal users and local actors. Cluster Made in Italy will coordinate training formats and evaluation questionnaires. GAL Pollino will support territorial participation, especially in rural and community contexts.

D 3.2.1 - Training programme and materials

Deliverable title	Target value	Delivery date
Training programme and materials	1	31/05/2028

Description

Training package covering Digital Twin use, scenario interpretation, prevention planning, communication procedures and preparedness exercises. Responsible: UNICAL and Cluster Made in Italy.

D 3.2.2 - Training and exercise completion report

Deliverable title	Target value	Delivery date
Training and exercise completion report	1	30/11/2028

Description

Report documenting training sessions, participants, feedback and learning outcomes. At least 90 people trained or actively involved by the final session. Responsible: Cluster Made in Italy with Urban Authorities.

A 3.3 - Smart wildfire-risk communication for target groups

Title: Smart wildfire-risk communication for target groups
Start date: 31/12/2027
End date: 31/12/2028

Activity description and Partners involved

This activity will design and test smart risk-communication procedures for differentiated target groups: residents, vulnerable people, farmers, visitors, local operators and civil-protection actors. Cluster Made in Italy will lead the design of messages, channels and formats. Castrovillari and Cassano will validate local relevance. GAL Pollino will support outreach to rural and community stakeholders. UNICAL will ensure technical accuracy and Calabria Region will check consistency with regional AIB communication.

D 3.3.1 - Smart risk-communication toolkit

Deliverable title: Smart risk-communication toolkit
Target value: 1
Delivery date: 31/08/2028

Description

Communication toolkit with messages, templates, target-group guidance and channels for wildfire prevention and preparedness. Responsible: Cluster Made in Italy.

D 3.3.2 - Communication testing report

Deliverable title: Communication testing report
Target value: 1
Delivery date: 31/12/2028

Description

Report on testing communication procedures with at least five target groups, including feedback, improvements and final recommendations. Responsible: Cluster Made in Italy with GAL Pollino and Urban Authorities.

A 3.4 - Impact monitoring, validation and learning

Title: Impact monitoring, validation and learning
Start date: 30/09/2027
End date: 28/02/2029

Activity description and Partners involved

This activity will monitor project results and validate whether the solution improves municipal capacity for wildfire prevention, preparedness and communication. Castrovillari will coordinate the monitoring process as MUA. UNICAL will measure technical and scenario-related indicators. Cluster Made in Italy will collect user feedback and training/communication evidence. Cassano will contribute municipal validation. Calabria Region will assess policy relevance and regional coherence.

D 3.4.1 - Impact monitoring framework

Deliverable title: Impact monitoring framework
Target value: 1
Delivery date: 31/12/2027

Description

Monitoring framework covering result indicators, data sources, responsibilities, baselines, target values, collection methods and reporting schedule. Responsible: Castrovillari with UNICAL.

D 3.4.2 - Final validation and learning report

Deliverable title	Target value	Delivery date
Final validation and learning report	1	29/02/2028

Description

Final report assessing results, indicator achievement, user feedback, lessons learned, limits, corrective actions and recommendations. Responsible: Castrovillari with all partners.

A 3.5 - Replication toolkit and transferability pathway

Title	Start date	End date
Replication toolkit and transferability pathway	31/12/2027	31/01/2029

Activity description and Partners involved

This activity will develop a replication toolkit for other wildfire-exposed municipalities and organise a structured transferability dialogue. It will not create an additional living lab or pilot territory. Cluster Made in Italy will coordinate the toolkit and transfer activities. UNICAL will define technical requirements. Calabria Region will support policy uptake and connection with regional strategies. Castrovillari and Cassano will share tested municipal lessons. GAL Pollino will support territorial transfer.

D 3.5.1 - Replication toolkit

Deliverable title	Target value	Delivery date
Replication toolkit	1	31/12/2027

Description

Toolkit including data-readiness checklist, governance model, functional requirements, communication protocol, maintenance guidance and replication steps. Responsible: Cluster Made in Italy with UNICAL.

D 3.5.2 - Transferability dialogue report

Deliverable title	Target value	Delivery date
Transferability dialogue report	1	31/01/2029

Description

Report documenting engagement with at least 10 external municipalities, institutional actors or territorial stakeholders and assessing replication conditions. Responsible: Cluster Made in Italy with Calabria Region.

A 3.6 - Policy uptake, sustainability and final dissemination

Title	Start date	End date
Policy uptake, sustainability and final dissemination	31/10/2027	31/01/2029

Activity description and Partners involved

This communication-related activity will consolidate project results for long-term uptake. Calabria Region will support policy alignment and regional capitalisation. Castrovillari and Cassano will define how outputs will be embedded in municipal routines. Cluster Made in Italy will

coordinate final dissemination and exploitation materials. UNICAL will provide technical handover guidance. GAL Pollino will support territorial dissemination and community uptake.

D 3.6.1 - Sustainability and handover plan

Deliverable title: Sustainability and handover plan
Target value: 1
Delivery date: 31/01/2029
Description: Plan defining post-project ownership, maintenance needs, use of outputs, municipal routines, technical handover and update responsibilities. Responsible: Castrovillari with UNICAL and Cluster Made in Italy.

D 3.6.2 - Policy uptake and final dissemination package

Deliverable title: Policy uptake and final dissemination package
Target value: 1
Delivery date: 31/01/2029
Description: Package including policy recommendations, final communication products, lessons learned, replication messages and evidence for regional and external uptake. Responsible: Cluster Made in Italy with Calabria Region.

Outputs

O.3.1 - Smart risk communication and preparedness protocol

Title: Smart risk communication and preparedness protocol
Target value: 1
Delivery date: 31/12/2028
Output indicator: New products and services created
Unit: new products/services
Description: Operational protocol and communication toolkit for using Digital Twin outputs in municipal preparedness and targeted wildfire-risk communication. Responsible: Cluster Made in Italy with Urban Authorities, UNICAL and Calabria Region.

O.3.2 - Trained municipal and local operator network

Title: Trained municipal and local operator network
Target value: 90
Delivery date: 30/11/2028
Output indicator: People supported (trained, upskilled, accompanied or assisted)
Unit: persons
Description: Network of municipal staff, civil-protection actors, local operators and stakeholders trained or actively involved in preparedness, Digital Twin use, scenario interpretation and communication procedures. Responsible: Cluster Made in Italy with UNICAL and Urban Authorities.

O.3.3 - Replication toolkit for wildfire-resilient municipalities

Title	Target value	Delivery date
Replication toolkit for wildfire-resilient municipalities	1	31/12/2028

Output indicator	Unit
New products and services created	new products/services

Description

Toolkit enabling other municipalities to assess and adapt the Twin4forestLife model, including data readiness, governance, technical requirements, communication protocol and maintenance guidance. Responsible: Cluster Made in Italy with UNICAL and Calabria Region.

O.3.4 - Impact validation and policy uptake package

Title	Target value	Delivery date
Impact validation and policy uptake package	1	28/02/2029

Output indicator	Unit
New products and services created	new products/services

Description

Package consolidating result indicators, evidence, lessons learned, validation findings, policy recommendations and transferability evidence. Responsible: Castrovillari and Cluster Made in Italy with all partners.

Investment(s)

Budget

Instructions

Depending on the selected **Cost Option** (in Part B "Partners"), each Partner should calculate their budget according to the following rules:

Option 1

Standard hourly rate per Member State for Staff costs, flat rate for Office and Administration, Travel and Accommodation; real cost for the rest of cost categories

- **Staff expenditures** are declared by Project Partners as direct staff costs on the basis of a simplified cost option method: a standard hourly rate per Member State. The hourly rate is standard for each Project Partner. The table of hourly rates per Member State is available here: **labour cost levels - hourly rate per Member State** . This table should be used by applicants for the Staff costs calculations, and it is based on the Eurostat data for Labour cost levels for each Member State. While calculating the staff costs, applicants should make sure to use the most up-to-date hourly rate available in the linked Eurostat data. Updating the calculations – should more recent data be published – will be possible during the Initiation Phase.
- **Office and Administration expenditures** are covered by an automatic flat rate of 15% of the reported Staff costs.
- **Travel and Accommodation expenditures** are covered by an automatic flat rate of 5% of the reported Staff costs.
- **External Expertise and Services, Equipment, and Infrastructure and Construction Works** expenditures are direct real costs.

Option 2

Standard hourly rate per Member State for Staff costs and 40% flat rate of Staff costs for other cost categories

- **Staff expenditures** are declared by Project Partners as direct staff costs on the basis of a simplified cost option method: a standard hourly rate per Member State. The hourly rate is standard for each Project Partner. The table of hourly rates per Member State is available here: **labour cost levels - hourly rate per Member State** . This table should be used by applicants for the Staff costs calculations, and it is based on the Eurostat data for Labour cost levels for each Member State. While calculating the staff costs, applicants should make sure to use the most up-to-date hourly rate available in the linked Eurostat data. Updating the calculations – should more recent data be published – will be possible during the Initiation Phase.
- **All other expenditures** under Office and Administration, Travel and Accommodation, External Expertise and Services, Equipment, Infrastructure and Construction Works cost categories are automatically calculated as a 40% flat rate on Staff costs. The beneficiary only needs to document expenditure related to Staff costs.

Option 3

20% flat rate for Staff costs, flat rate for Office and Administration, Travel and Accommodation; real cost for the rest of cost categories

- **Staff expenditure** is automatically calculated as a 20% flat rate of the project direct costs (External Expertise and Services + Equipment + Infrastructure & Construction Works).
- **Office and Administration expenditure** is covered by a flat rate of 15% of the reported Staff costs.
- **Travel and Accommodation expenditure** is covered by a flat rate of 5% of the reported Staff costs.
- **External Expertise and Services, Equipment, and Infrastructure and Construction Works** expenditures are direct real costs.

Applicants are advised to refer to **Chapter 6 of the EUI-IA Guidance** while developing their project budget.

Project Partner 1 - Munipality of Castrovillari

Budget Line	Preparation Lump Sum	Initiation Phase Lump Sum	Allocated Budget Description	Allocated Budget Amount	Closure Lump Sum	Total
Staff (€)	0.00	0.00	Municipal project coordinator, technical lead for the Castrovillari living lab, civil-protection liaison officer, administrative and financial management officer, GIS/data officer supporting baseline and pilot activities. Costs cover salaries/fees for approx. 24 months across WP1, WP2 and WP3.	250,000.00	0.00	250,000.00
Office and Administration (€)	0.00	0.00	N/A	37,500.00	0.00	37,500.00
Travel and Accommodation (€)	0.00	0.00	N/A	12,500.00	0.00	12,500.00
External Expertise and Services (€)	10,000.00	30,000.00	External technical services for Digital Twin integration, GIS and spatial data processing, drone/sensor data acquisition and processing, scenario development support, legal/administrative procurement support, communication services and event organisation for the Castrovillari living lab and transferability activities. Preparation lump sum: €10,000; Initiation phase lump sum: €30,000; Closure lump sum: €7,500.	177,500.00	7,500.00	225,000.00
Equipment (€)	0.00	0.00	No needed	0.00	0.00	0.00
Infrastructure and Works (€)	0.00	0.00	No needed	0.00	0.00	0.00
Total (€)	**10,000.00**	**30,000.00**		**477,500.00**	**7,500.00**	**525,000.00**

Project Partner 2 - Municipality of Cassano allo Ionio

Budget Line	Allocated Budget Description	Allocated Budget Amount
Staff (€)	Municipal technical officer and civil-protection liaison for the Cassano living lab, administrative and financial contact for WP activities. Costs cover salaries/fees for the implementation of Cassano's pilot activities across WP1, WP2 and WP3.	180,000.00

Budget Line	Allocated Budget Description	Allocated Budget Amount
Office and Administration (€)	N/A	27,000.00
Travel and Accommodation (€)	N/A	9,000.00
External Expertise and Services (€)	External services supporting the Cassano living lab: Digital Twin pilot testing support, scenario validation, emergency-area and access-route analysis, Civil Protection Plan integration, stakeholder engagement activities, communication and risk-communication testing services.	159,000.00
Equipment (€)	Hardware for local data access and Digital Twin operational use in the Cassano living lab: workstations, displays and connectivity equipment required for pilot testing, scenario visualisation and municipal preparedness activities.	25,000.00
Infrastructure and Works (€)	No needed	0.00
Total (€)		400,000.00

Project Partner 3 - Cluster Tecnologico Nazionale “Made in Italy”

Budget Line	Allocated Budget Description	Allocated Budget Amount
Staff (€)	ommunication and dissemination manager, stakeholder engagement coordinator, project administration support. Costs cover personnel directly responsible for communication strategy, stakeholder-oriented material production, dissemination events, replication toolkit development and policy uptake activities across WP1, WP2 and WP3.	165,000.00
Office and Administration (€)	N/A	24,750.00
Travel and Accommodation (€)	N/A	8,250.00
External Expertise and Services (€)	External services for visual identity and communication materials, web presence, dissemination products, co-creation format design, transferability workshop organisation, stakeholder-oriented tools for risk communication and replication, content production for EUI visibility obligations.	152,000.00
Equipment (€)	No needed	0.00
Infrastructure and Works (€)	No needed	0.00

Budget Line	Allocated Budget Description	Allocated Budget Amount
Total (€)		350,000.00

Project Partner 4 - University of Calabria

Budget Line	Allocated Budget Description	Allocated Budget Amount
Staff (€)	Research and technical team responsible for Digital Twin design, development and validation: Digital Twin architect, data integration specialists, GIS/remote sensing experts, climate and wildfire-risk modelling researchers, software developers, scenario-design and validation officers. Costs cover personnel fees/salaries for the full project duration (WP1–WP3).	386,400.00
Office and Administration (€)	N/A	57,960.00
Travel and Accommodation (€)	N/A	19,320.00
External Expertise and Services (€)	External services for specific technical components: specialised software licences or services, cloud/server infrastructure for Digital Twin hosting and data processing, third-party data acquisition or processing services not covered by internal resources.	36,320.00
Equipment (€)	No needed	0.00
Infrastructure and Works (€)	No needed	0.00
Total (€)		500,000.00

Project Partner 5 - Local Action Group Pollino

Budget Line	Allocated Budget Description	Allocated Budget Amount
Staff (€)	Territorial engagement coordinator, rural community liaison officer, administrative and financial support contact. Costs cover personnel responsible for stakeholder mobilisation, local outreach, participatory workshops, community validation activities and support to transferability in the Castrovillari/Pollino area across WP1–WP3.	145,000.00
Office and Administration (€)	N/A	21,750.00

Budget Line	Allocated Budget Description	Allocated Budget Amount
Travel and Accommodation (€)	N/A	7,250.00
External Expertise and Services (€)	External services for local engagement events, participatory workshop facilitation, rural and environmental stakeholder outreach, community communication materials, local validation activities and territorial transferability support in the GAL Pollino area.	126,000.00
Equipment (€)	No needed	0.00
Infrastructure and Works (€)	No needed	0.00
Total (€)		300,000.00

Project Partner 6 - Calabria Region

Budget Line	Allocated Budget Description	Allocated Budget Amount
Staff (€)	Regional AIB and civil-protection liaison officers, Digital Twin ecosystem coordination staff, policy alignment and data-governance contact. Costs cover regional personnel involved in validation, regional data sharing, interoperability coordination and policy uptake activities across WP1–WP3.	85,000.00
Office and Administration (€)	N/A	12,750.00
Travel and Accommodation (€)	N/A	4,250.00
External Expertise and Services (€)	External services for regional wildfire data processing, Digital Twin interoperability support, AIB framework integration, monitoring and evaluation, policy uptake events and coordination with the regional Control Room. Consultancy services will support the Region in implementing wildfire prevention measures and risk-reduction interventions on the territory, aligned with the regional AIB framework and civil-protection priorities.	293,000.00
Equipment (€)	Hardware for regional-municipal data exchange: servers, connectivity equipment and field devices supporting data sharing between the regional AIB/Digital Twin systems and the municipal pilot environments in Castrovillari and Cassano all'Ionio.	30,000.00
Infrastructure and Works (€)	No needed	0.00
Total (€)		425,000.00

Total Budget

	Staff	Office and Administration	Travel and Accommodation	External Expertise and Services	Equipment	Infrastructure and Works	Total
Amount (€)	1,211,400.00	181,710.00	60,570.00	991,320.00	55,000.00	0.00	2,500,000.00

Indicative Budget Breakdown Per Year

Year	Preparation & Initiation Phase	2027	2028	2029	Closure	Total
% Budget	1.60	35.00	58.10	5.00	0.30	100.00
Amount (€)	40,000.00	875,000.00	1,452,500.00	125,000.00	7,500.00	2,500,000.00

Indicative Budget Breakdown Per Work Package

WorkPackage	Preparation & Initiation Phase	WP 1 - Baseline, governance and Digital Twin architecture	WP 2 - Digital Twin deployment and wildfire-risk pilot actions	WP 3 - Operational preparedness, uptake and replication	Closure	Total
% Budget	1.60	15.00	70.00	13.10	0.30	100.00
Amount (€)	40,000.00	375,000.00	1,750,000.00	327,500.00	7,500.00	2,500,000.00

Part E - Project Budget

E.1 Project Budget Co-Financing Source (Fund) - Breakdown per Partner

Partner	Country	ERDF co-financing (maximum 80%)		Contribution (minimum 20%)			Total 100% TEC (total eligible costs)	
		EUR	ERDF rate	Public	Private	Total	Budget	% of project budget
PP 1 - Munipality of Castrovillari	IT	420,000.00	80.00%	105,000.00	0.00	105,000.00	525,000.00	21.00%
PP 2 - Municipality of Cassano allo Ionio	IT	320,000.00	80.00%	80,000.00	0.00	80,000.00	400,000.00	16.00%
PP 3 - Cluster Tecnologico Nazionale “Made in Italy”	IT	280,000.00	80.00%	70,000.00	0.00	70,000.00	350,000.00	14.00%
PP 4 - University of Calabria	IT	400,000.00	80.00%	100,000.00	0.00	100,000.00	500,000.00	20.00%
PP 5 - Local Action Group Pollino	IT	240,000.00	80.00%	0.00	60,000.00	60,000.00	300,000.00	12.00%
PP 6 - Calabria Region	IT	340,000.00	80.00%	85,000.00	0.00	85,000.00	425,000.00	17.00%
Total (€)		**2,000,000.00**	**80%**	**440,000.00**	**60,000.00**	**500,000.00**	**2,500,000.00**	**100.00%**

E.2 Project Budget - Overview per Partner/ per Budget Line

Partner	Staff costs	Office and administration	Travel and accommodation	External expertise and services	Equipment	Infrastructure and works	Total
PP 1	250,000.00	37,500.00	12,500.00	225,000.00	0.00	0.00	525,000.00
PP 2	180,000.00	27,000.00	9,000.00	159,000.00	25,000.00	0.00	400,000.00
PP 3	165,000.00	24,750.00	8,250.00	152,000.00	0.00	0.00	350,000.00
PP 4	386,400.00	57,960.00	19,320.00	36,320.00	0.00	0.00	500,000.00
PP 5	145,000.00	21,750.00	7,250.00	126,000.00	0.00	0.00	300,000.00
PP 6	85,000.00	12,750.00	4,250.00	293,000.00	30,000.00	0.00	425,000.00
Total (€)	**1,211,400.00**	**181,710.00**	**60,570.00**	**991,320.00**	**55,000.00**	**0.00**	**2,500,000.00**
% of total budget	**48.46%**	**7.27%**	**2.42%**	**39.65%**	**2.20%**	**0.00%**	**100.00%**

Part F - Partners contribution

Source(s) of Contribution

Total Contribution	500,000.00
Total Contribution Target	500,000.00

PP1 - Munipality of Castrovillari

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
Comune di Castrovillari	Public	100.00%	105,000.00	cash	In-kind contribution in the form of staff time, office space, municipal data, GIS resources and civil-protection expertise made available by Castrovillari for project implementation.
Total (€)		100%	105,000.00		Contribution Target 105,000.00

PP2 - Municipality of Cassano allo Ionio

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
Comune di Cassano allo Ionio	Public	100.00%	80,000.00	cash	Own municipal funds allocated as co-financing contribution, in accordance with the partnership agreement and EUI-IA Call 4 co-financing requirements.
Total (€)		100%	80,000.00		Contribution Target 80,000.00

PP3 - Cluster Tecnologico Nazionale “Made in Italy”

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
Cluster MINIT	Public	100.00%	70,000.00	cash	Own institutional funds allocated as co-financing contribution, in accordance with the partnership
Total (€)		100%	70,000.00		Contribution Target 70,000.00

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
					agreement and EUI-IA Call 4 co-financing requirements.
Total (€)		100%	70,000.00		Contribution Target 70,000.00

PP4 - University of Calabria

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
Università della Calabria	Public	100.00%	100,000.00	cash	Own institutional funds allocated as co-financing contribution, in accordance with the partnership agreement and EUI-IA Call 4 co-financing requirements.
Total (€)		100%	100,000.00		Contribution Target 100,000.00

PP5 - Local Action Group Pollino

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
GAL Pollino	Private	100.00%	60,000.00	cash	Own institutional funds allocated as co-financing contribution, in accordance with the partnership agreement and EUI-IA Call 4 co-financing requirements.
Total (€)		100%	60,000.00		Contribution Target 60,000.00

PP6 - Calabria Region

Name of Organisation/ Source of Contribution	Legal Status	% of Total Partner Contribution	Amount (€)	Cash or In-kind Contribution	Comment
Regione Calabria	Public	100.00%	85,000.00	cash	Own regional funds allocated as co-financing contribution, in accordance with the partnership agreement and EUI-IA Call 4 co-financing requirements.
Total (€)		100%	85,000.00		Contribution Target 85,000.00

Part G - Risk Management

G.1 Risks at project strategy level

The main strategic risk is that the solution may be perceived as duplicating existing regional AIB, Control Room or Digital Twin systems. This will be managed by clearly positioning Twin4forestLife as the municipal operational layer of the regional ecosystem — not replacing firefighting or emergency-command structures, but testing how regional intelligence can be used by municipalities for prevention, preparedness, access planning, water logistics and risk communication.
A second risk concerns political and administrative continuity. This will be mitigated by ensuring strong ownership by both Urban Authorities, embedding outputs into municipal prevention and civil-protection routines, and defining a sustainability and handover plan in WP3.
A third risk is limited stakeholder uptake if the Digital Twin is perceived as too technical. The project will address this through co-creation, training, user-friendly dashboards, practical scenarios and smart communication tools tailored to all target groups.
A fourth risk concerns data availability and interoperability. This will be mitigated in WP1 through a data inventory, governance protocol, clear access rules and realistic MVP requirements, avoiding dependence on unavailable or restricted data.
Finally, there is a risk of over-ambition. The project deliberately focuses on two municipalities and two living labs, ensuring the solution remains feasible and testable within 24 months.

G.2 Risks at project implementation level

Project management capacity risks

Risk	Description	Properties	Actions to mitigate the risk
Structures and procedures	The project involves municipalities, university, regional authority and territorial/communication partners. Delays may occur if roles, deadlines and validation procedures are not clear.	Likelihood: Possible Impact: Serious	Establish Steering Committee, WP leads, monthly meetings, shared calendar, risk tracker, reporting templates and clear validation workflow from project start.
Project Partnership	Some partners may have limited availability during key phases, causing delays in data provision, validation, communication or stakeholder engagement.	Likelihood: Possible Impact: Minor	Confirm operational contacts, define partner role matrix in WP1, monitor engagement monthly and activate corrective actions through the MUA and Steering Committee.
Financial flows	Partners may face delays in expenditure, procurement or internal accounting, affecting budget absorption and reporting.	Likelihood: Possible Impact: Serious	MUA will coordinate financial reporting, quarterly budget checks, expenditure forecasts, evidence collection and early-warning procedures for underspending or delays.
Political environment	Local elections or changes in administrative priorities could reduce political attention to the project or slow decision-making.	Likelihood: Possible Impact: Minor	Anchor the project in municipal safety, civil-protection and climate-adaptation responsibilities; involve technical offices and

Risk	Description	Properties	Actions to mitigate the risk
			formal governance structures, not only political representatives.
Limited staff availability in small municipalities	Municipal staff may be overloaded by ordinary duties and emergency-management responsibilities, reducing time available for project testing.	Likelihood Possible Impact Serious	Plan realistic testing sessions, provide concise tools, integrate activities into existing municipal routines and avoid excessive administrative burden on local staff.

Work Plan implementation risks

Risk	Description	Properties	Actions to mitigate the risk
Legal framework and regulations regarding innovation (local, national, EU level)	Some datasets may be restricted, incomplete or subject to privacy, cybersecurity or institutional-access rules.	Likelihood Possible Impact Serious	WP1 will define a data inventory, access rules, GDPR checks, anonymisation where needed and a realistic MVP based only on legally usable data.
Delays in the delivery of project activities	Delays in baseline, data architecture or MVP development could reduce time for pilot testing, training and validation.	Likelihood Probable Impact Serious	Use phased delivery: minimum viable datasets first, early prototype by M10, parallel preparation of WP3, monthly technical reviews and corrective actions by Steering Committee.
Participatory approach	Residents, farmers, operators or volunteers may show limited interest if the Digital Twin is perceived as technical or distant from daily needs.	Likelihood Possible Impact Serious	Use practical scenarios, local meetings, targeted messages, GAL Pollino outreach, municipal communication channels and feedback formats adapted to each target group.
Over-complexity of the Digital Twin	The solution could become too complex for municipal users or too dependent on advanced technical support.	Likelihood Probable Impact Serious	Design user-centred MVP, prioritise essential functions, provide training, simplify dashboards, test usability with municipal staff and define a light maintenance model.

Investments risks

Risk	Description	Properties	Actions to mitigate the risk
(Public)procurement	Procurement of software, integration services, hosting or	Likelihood	Prepare technical specifications early in WP1, use market

Risk	Description	Properties	Actions to mitigate the risk
procedure	limited field equipment may take longer than planned or receive weak market response.	Probable Impact Serious	consultation where allowed, split lots if useful, apply transparent procurement rules and monitor procurement milestones.
Ownership & location of the site	Some minor monitoring or communication elements may require site checks or municipal authorisation before installation.	Likelihood Possible Impact Minor	Prioritise municipal-owned or controlled sites, verify authorisations in WP1/WP2, avoid installations requiring complex permits, and document ownership and access rights.
Delivery of investments	Technical integration, hosting configuration, data cleaning or equipment supply may delay MVP testing.	Likelihood Probable Impact Serious	Adopt modular deployment, test with minimum viable datasets, maintain backup suppliers where possible, and separate essential MVP functions from optional enhancements.
Obsolescence of technical solutions	Digital Twin, sensor or dashboard technologies may evolve during the project, risking reduced relevance of selected tools.	Likelihood Remote Impact Minor	Use open, modular and interoperable architecture; avoid vendor lock-in; prioritise data governance, use cases and municipal workflows over specific proprietary technologies.
Long-term maintenance burden	Municipalities may struggle to maintain advanced technical components after project end.	Likelihood Probable Impact Serious	Define a light maintenance model, handover package, technical documentation, staff training, municipal data-update routines and possible regional alignment/support mechanisms.

Part H - Confirmation

(Main) Urban Authority confirmation and signature

(Main) Urban Authority Munipality of Castrovillari

By signing the application form the (Main) Urban Authority hereby confirms that:

1. **The information provided in the Application Form is accurate and true to the best knowledge of the Main Urban Authority.**
2. **The Urban Authorities involved in this project proposal are not involved in other proposals submitted to the European Urban Initiative – Innovative Actions as part of this current Call for Proposals.**
3. **The Main Urban Authority and Project Partners listed in the Application Form:**
   a. **are committed to participate in the action,**
   b. **have stable and sufficient sources of funding to maintain the activities throughout the action and to provide any counterpart funding necessary,**
   c. **have or will have the necessary resources needed to implement the action.**
4. **The Main Urban Authority and the Project Partners commit to comply with the eligibility criteria and all other conditions set out in the Call for Proposals conditions for the entire duration of the project.**
5. **The Main Urban Authority and the Project Partners will act according to the provisions of the relevant national and EU legislation and policies (especially regarding structural funds, public procurement, state aid, environment and equal opportunities) as well as the specific provisions of the European Urban Initiative.**
6. **The Main Urban Authority and the Project Partners are NOT subject to an administrative sanction (i.e. exclusion or financial penalty decision).**
7. **The Main Urban Authority and the Project Partners (or persons with unlimited liability for debts) are NOT in one of the following exclusion situations :**
   a. **bankrupt, being wound up, having the affairs administered by the courts, entered into an arrangement with creditors, suspended business activities or subject to any other similar proceedings or procedures**
   b. **in breach of social security or tax obligations**
8. **The Main Urban Authority and the Project Partners (or persons having powers of representation, decision-making or control, beneficial owners or persons who are essential for the award/implementation of the action) are NOT in one of the following exclusion situations:**
   a. **guilty of grave professional misconduct,**
   b. **committed fraud, corruption, links to a criminal organisation, money laundering, terrorism-related crimes (including terrorism financing), child labour or human trafficking,**
   c. **shown significant deficiencies in complying with main obligations under an EU procurement contract, grant agreement, prize, expert contract, or similar,**
   d. **guilty of irregularities within the meaning of Article 1(2) of Regulation No 2988/95,**
   e. **created under a different jurisdiction with the intent to circumvent fiscal, social or other legal obligations in the country of origin (including creation of another entity with this purpose).**
9. **The Main Urban Authority and the Project Partners are NOT subject to a conflict of interest in connection with this grant and will notify — without delay — any situation which could give rise to a conflict of interests.**
10. **The Main Urban Authority and the Project Partners neither in whole nor in part have or will receive any other complementary EU funding (except for the funding indicated in this Application Form) during the whole duration of the project.**
11. **General information about this project can be used by the European Urban Initiative to liaise with national and regional authorities in charge of implementation of operational programmes funded by the European**

Structural and Investment Funds.

Forename, Surname	Fedele Enrico L' Avena	**Date**	12/06/2026
Position	Urban Planning Officer, delegated by the Mayor of Castrovillari	**Place**	Castrovillari
Authorized signature of (Main) Urban Authority			